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Huang et al.

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(54) **IMAGE CAPTURING LENS ASSEMBLY,
IMAGE CAPTURING DEVICE AND MOBILE
TERMINAL**

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(71) Applicant: **LARGAN Precision Co., Ltd.,**
Taichung (TW)

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(72) Inventors: **Hsin-Hsuan Huang**, Taichung (TW);
Tsung-Han Tsai, Taichung (TW)

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(73) Assignee: **LARGAN PRECISION CO., LTD.,**
Taichung (TW)

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Primary Examiner — David N Spector

(74) *Attorney, Agent, or Firm* — Locke Lord LLP; Tim
Ting kang Xia, Esq.

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(57) **ABSTRACT**

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G02B 13/00 (2006.01)
G02B 3/04 (2006.01)
H04N 5/225 (2006.01)

(52) **U.S. Cl.**

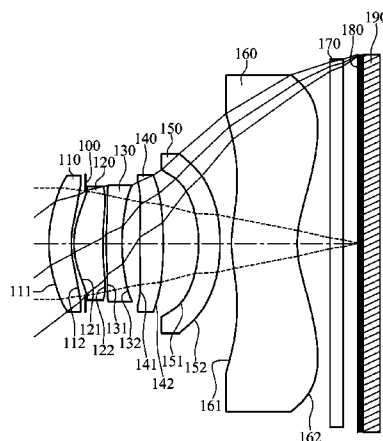
CPC **G02B 13/0045** (2013.01); **G02B 3/04**
(2013.01); **H04N 5/225** (2013.01)

(58) **Field of Classification Search**

CPC G02B 13/0045; G02B 9/62
USPC 359/713
See application file for complete search history.

An image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with negative refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element has refractive power. The fifth lens element has negative refractive power, wherein both of the surfaces thereof are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The image capturing lens assembly has a total of six lens elements with refractive power.

25 Claims, 21 Drawing Sheets



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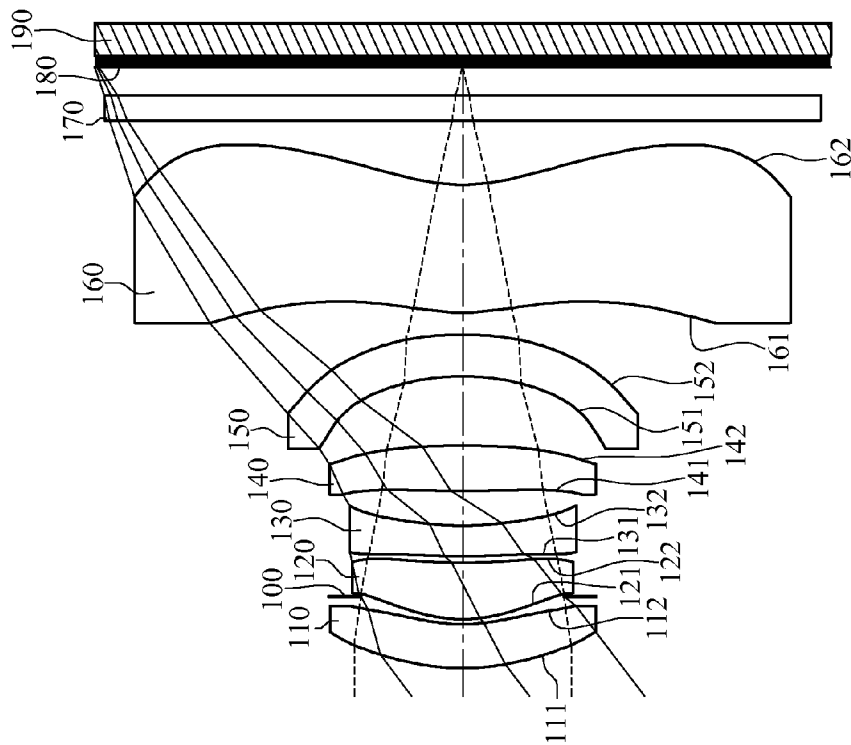
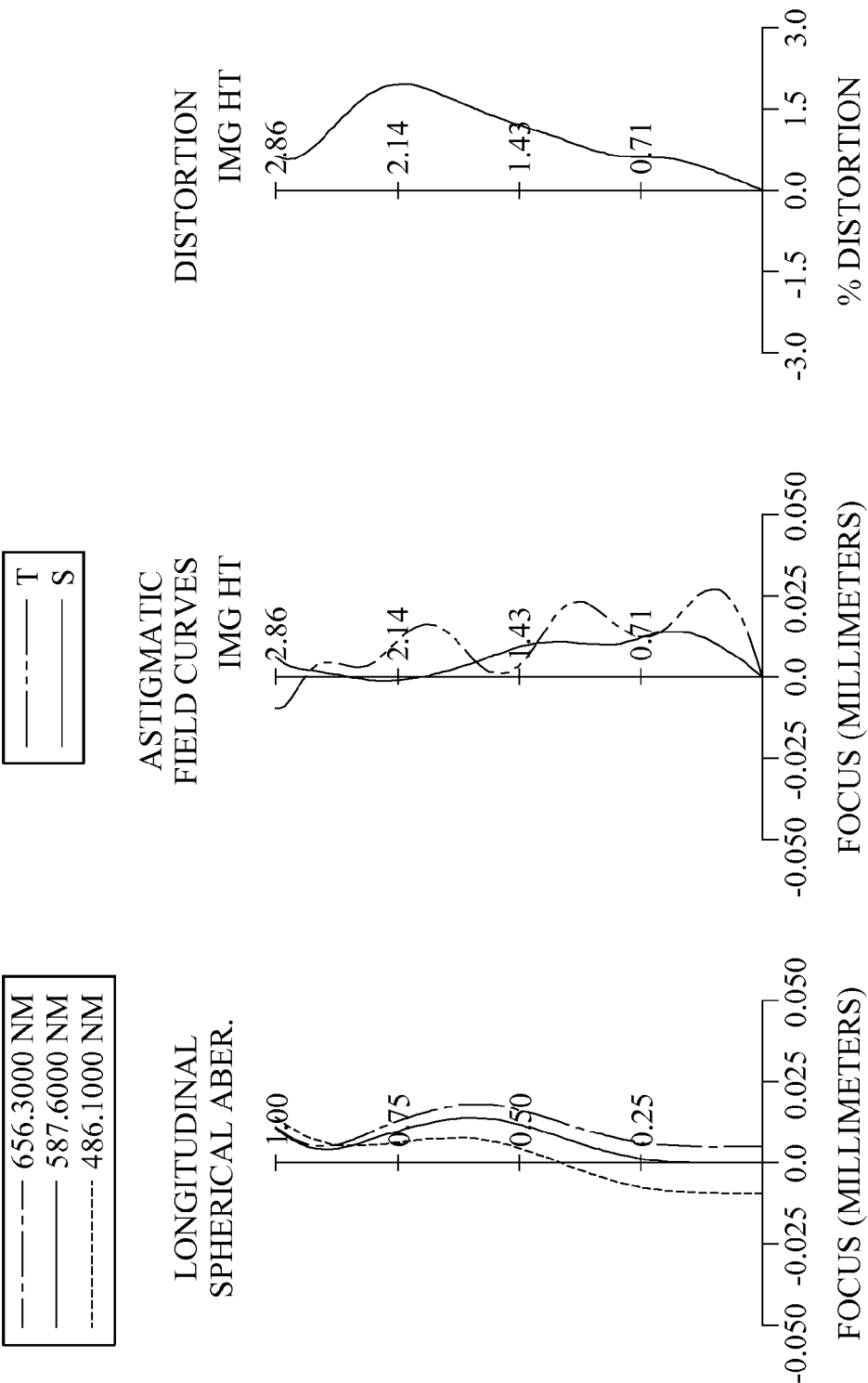


FIG.1



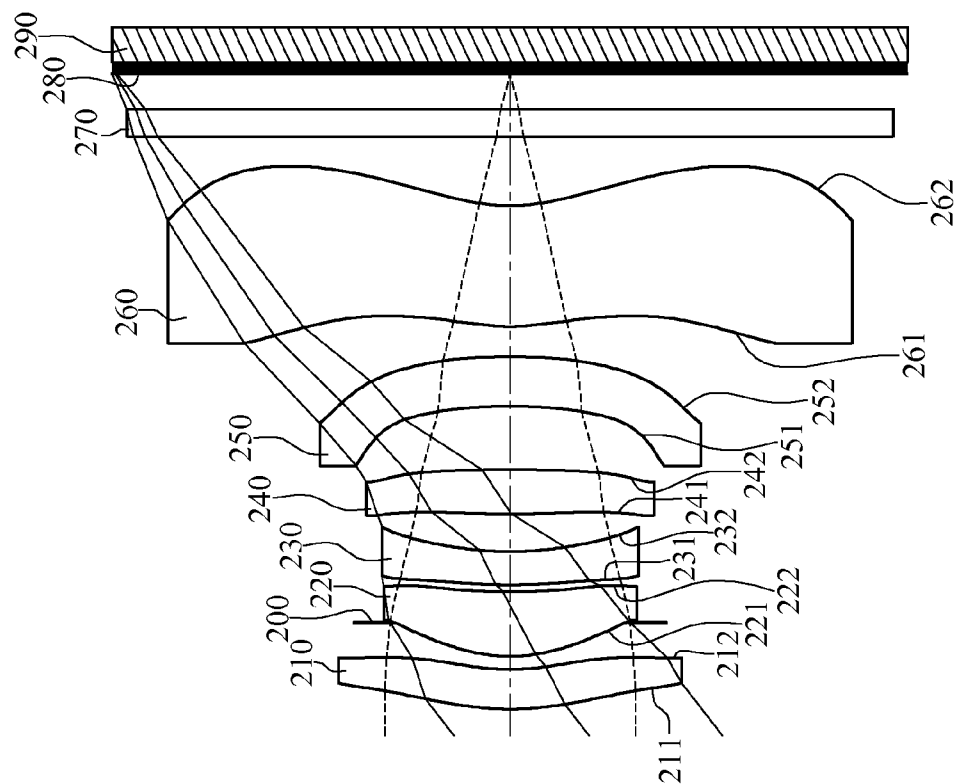
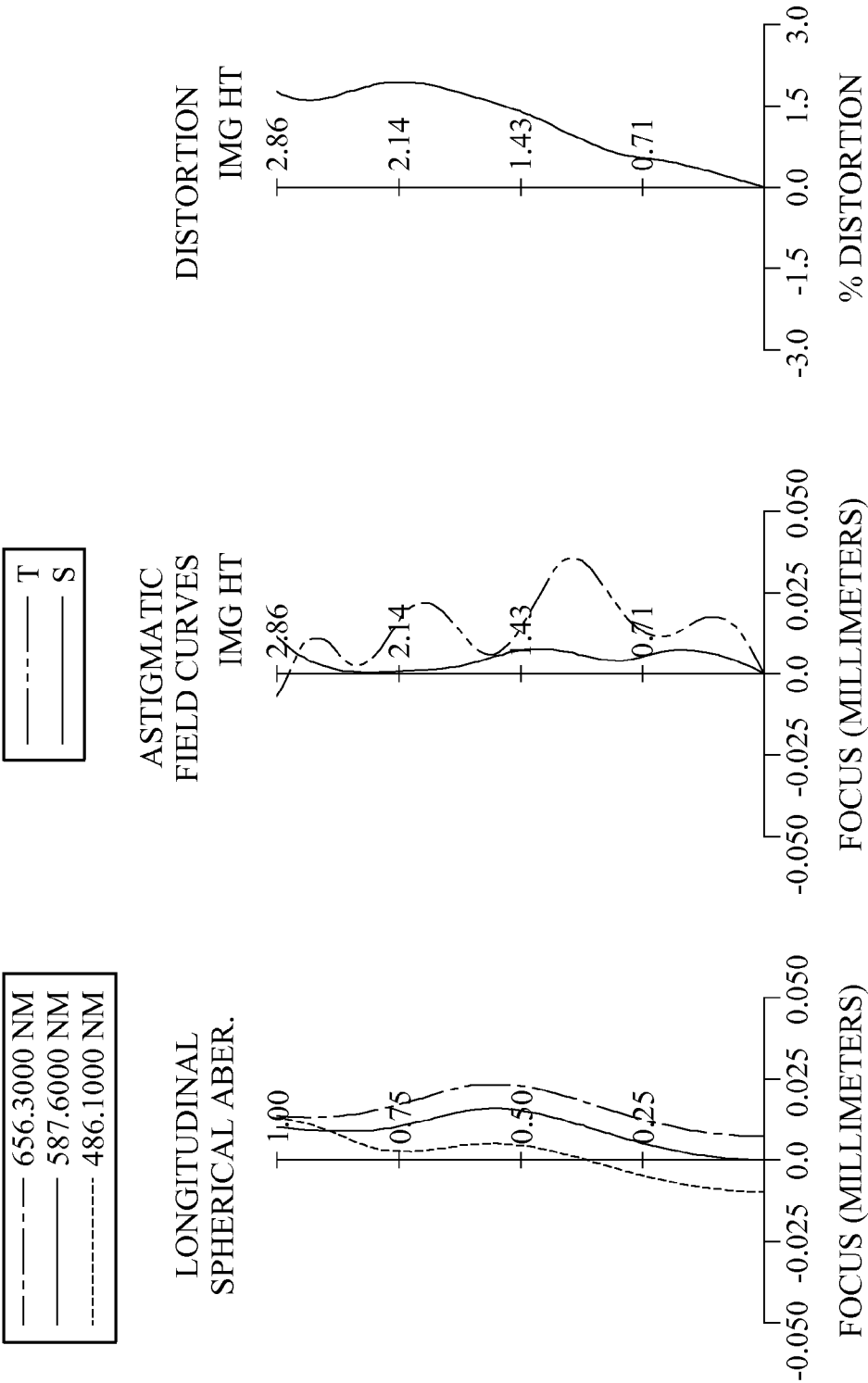


FIG.3



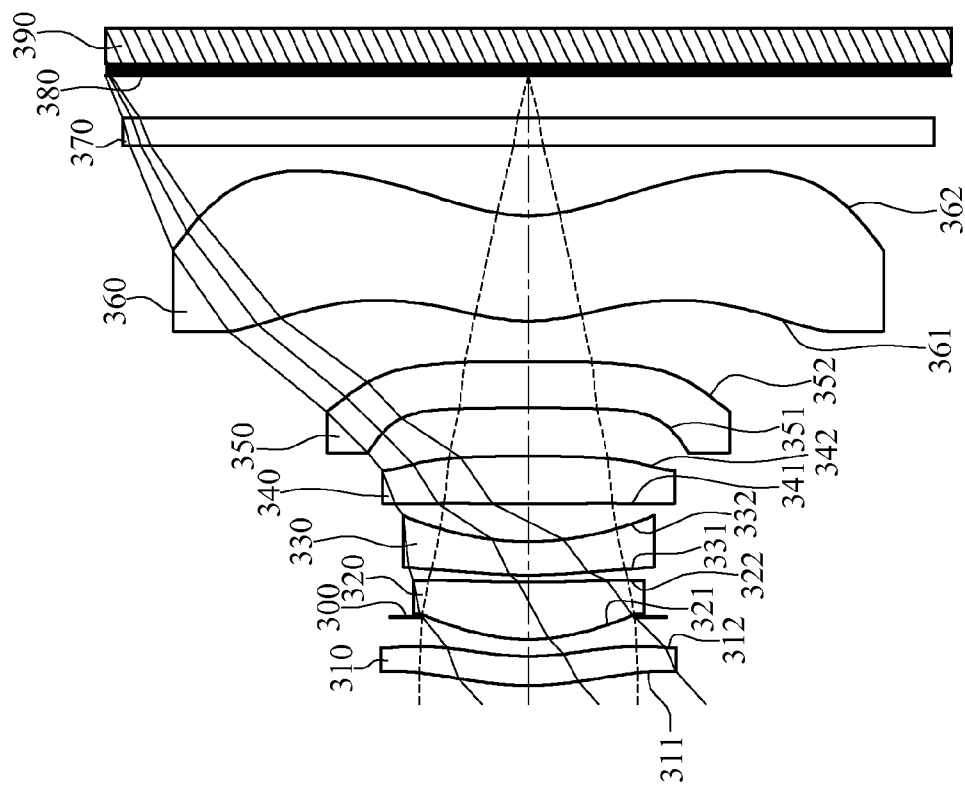


FIG.5

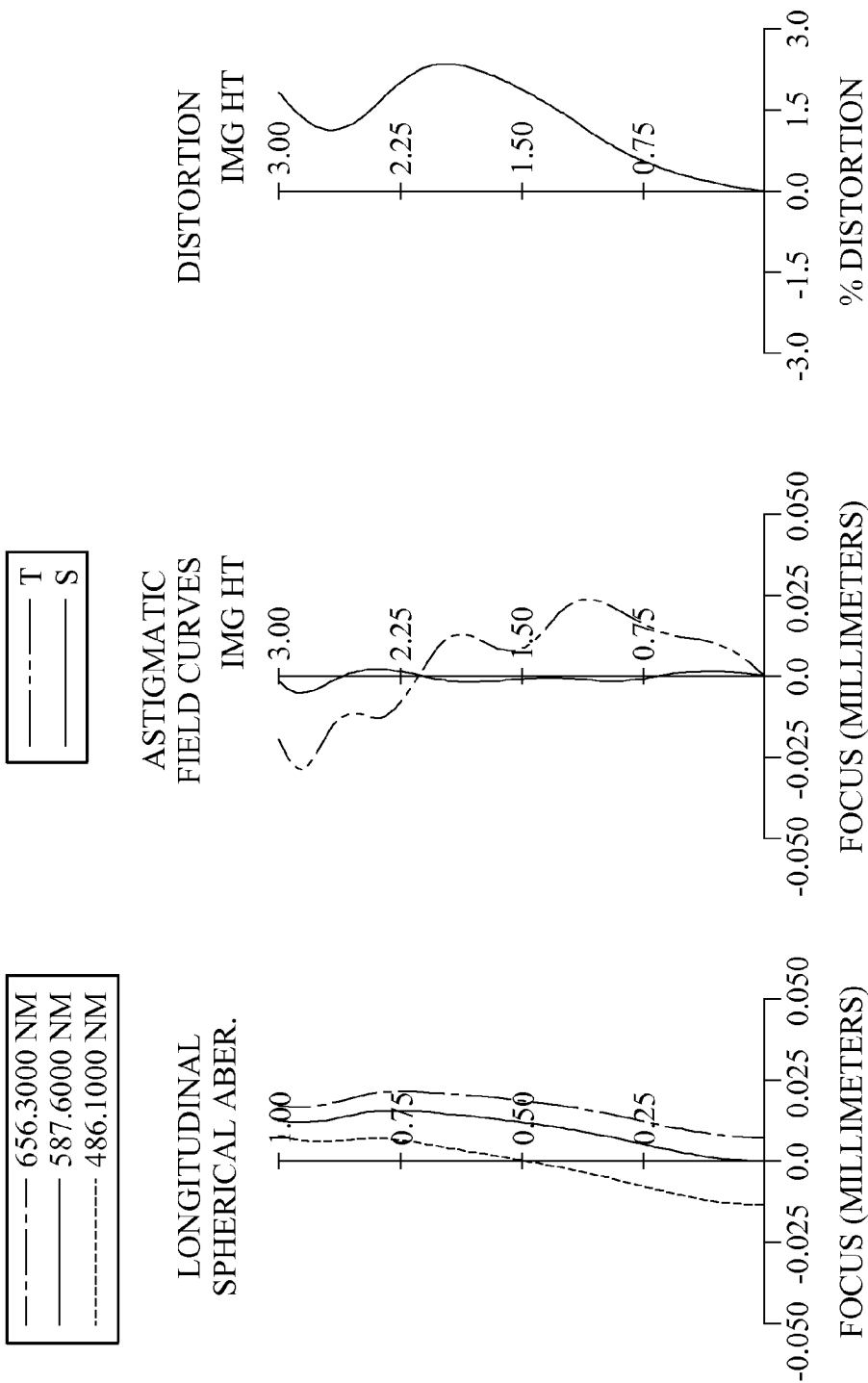


FIG.6

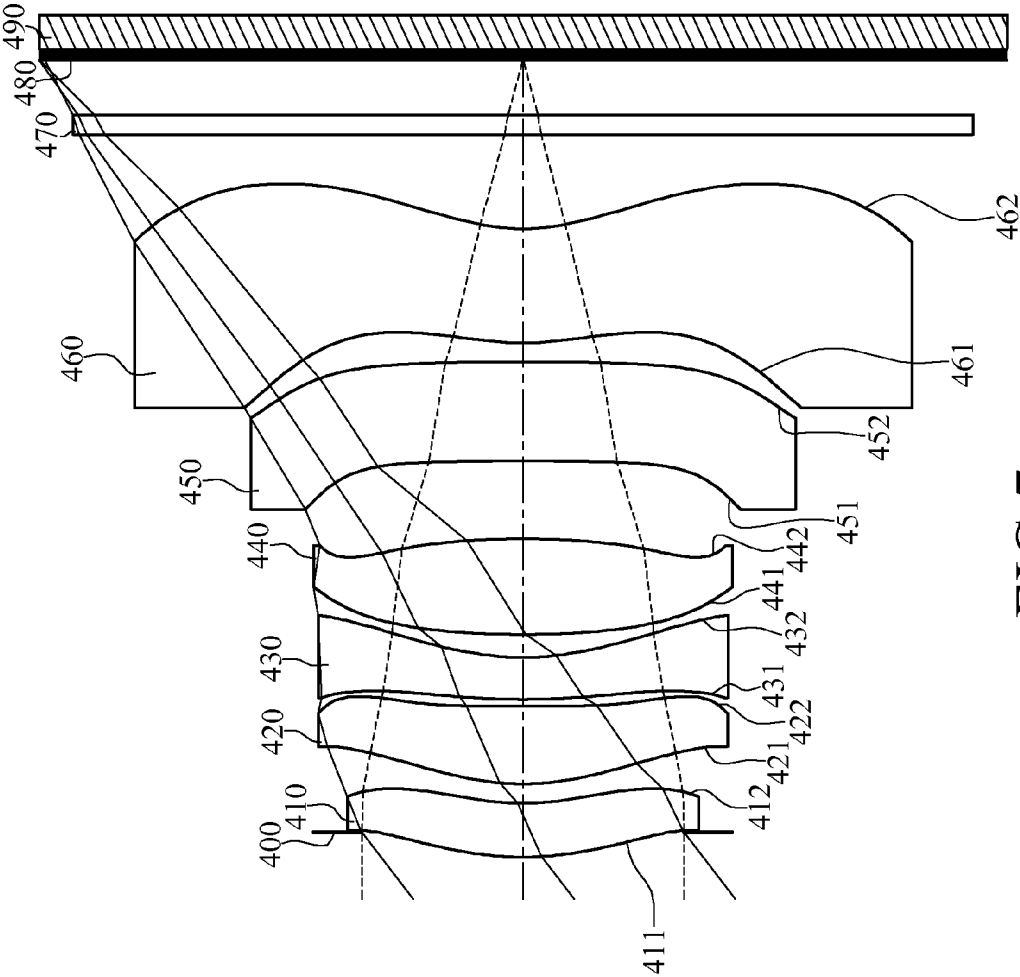


FIG. 7

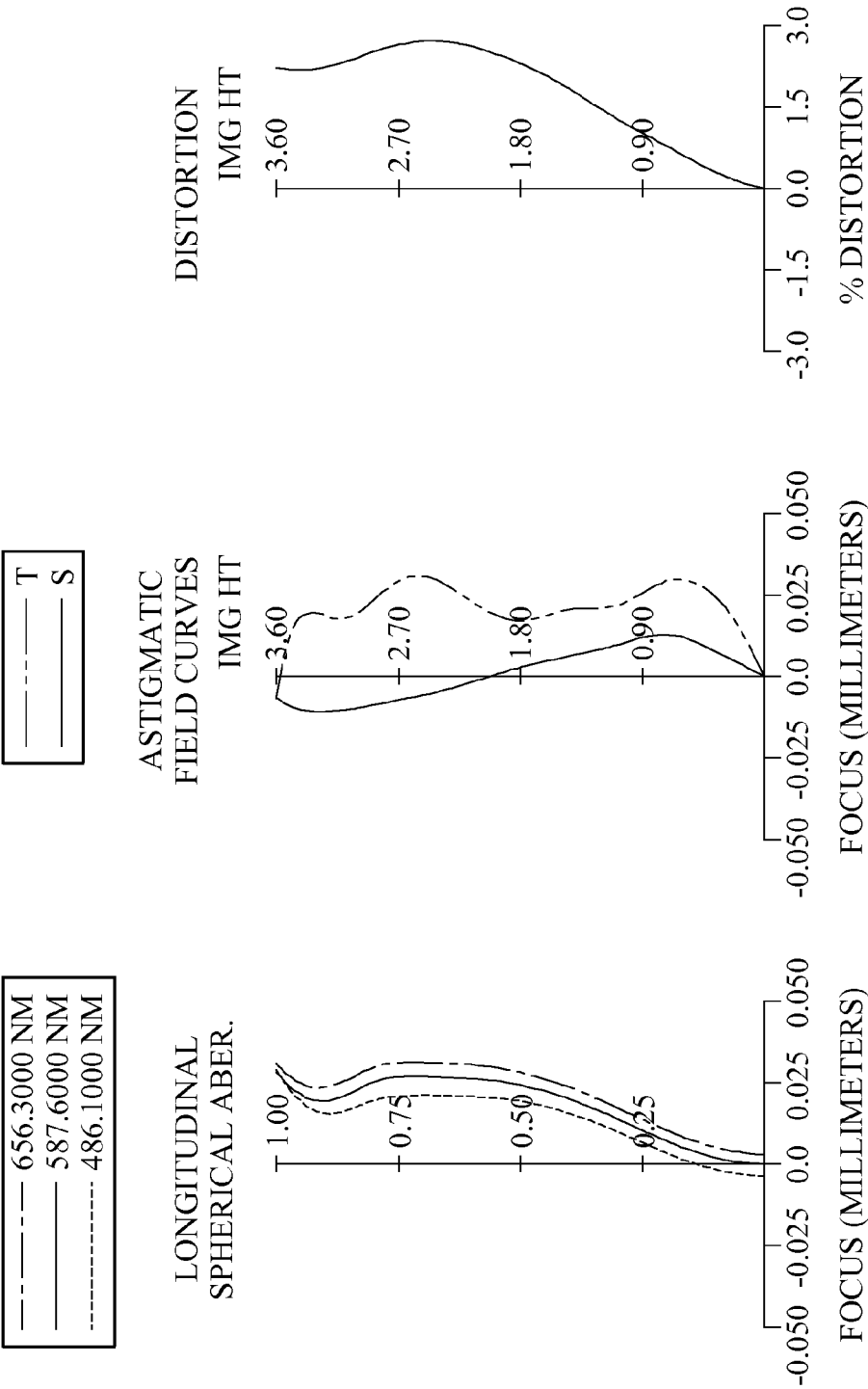


FIG.8

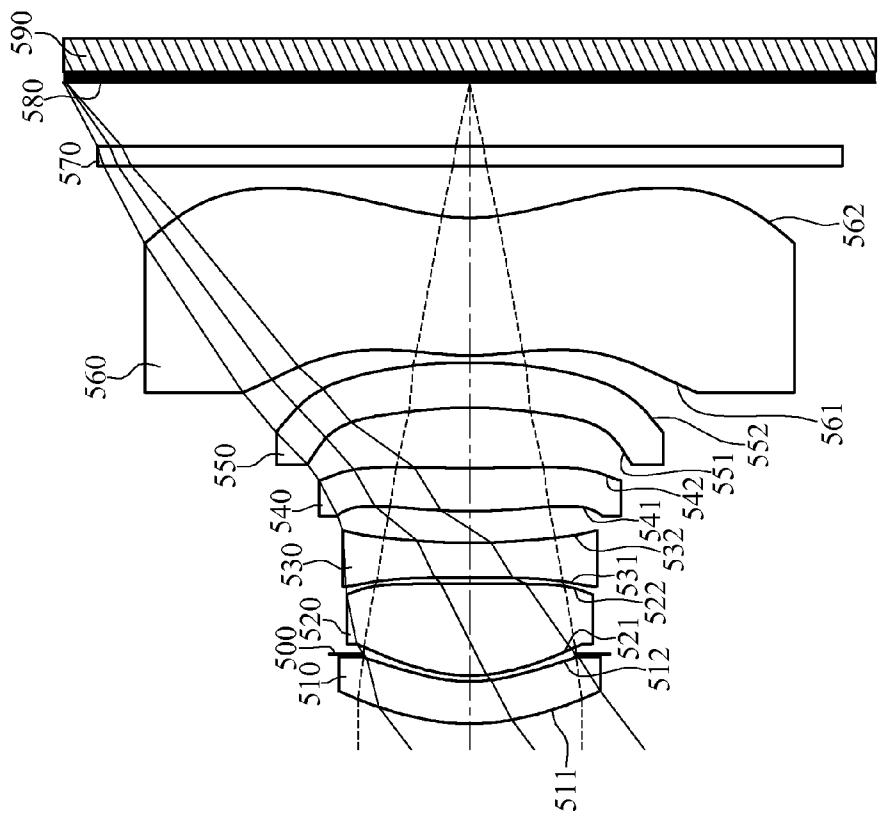
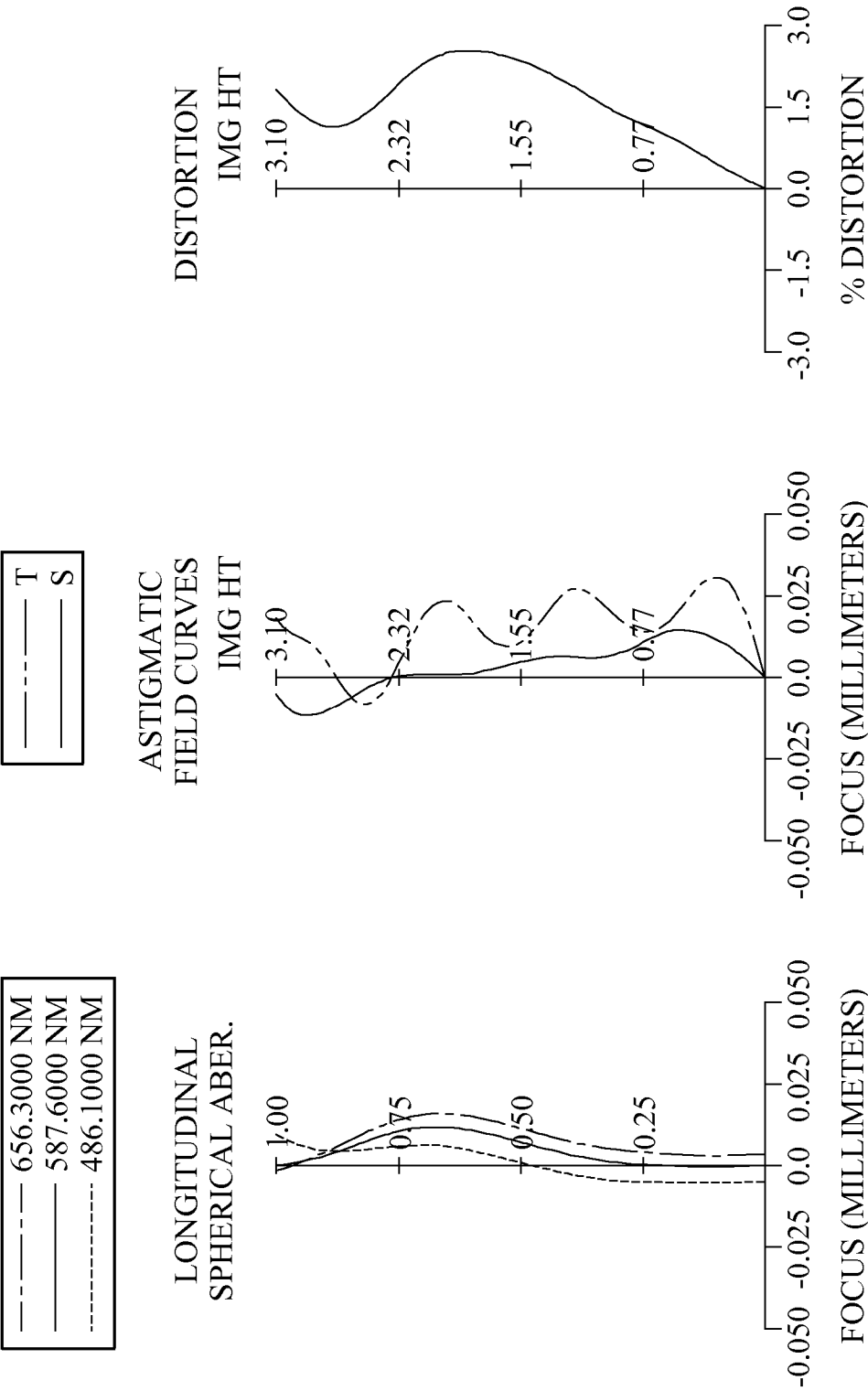


FIG.9



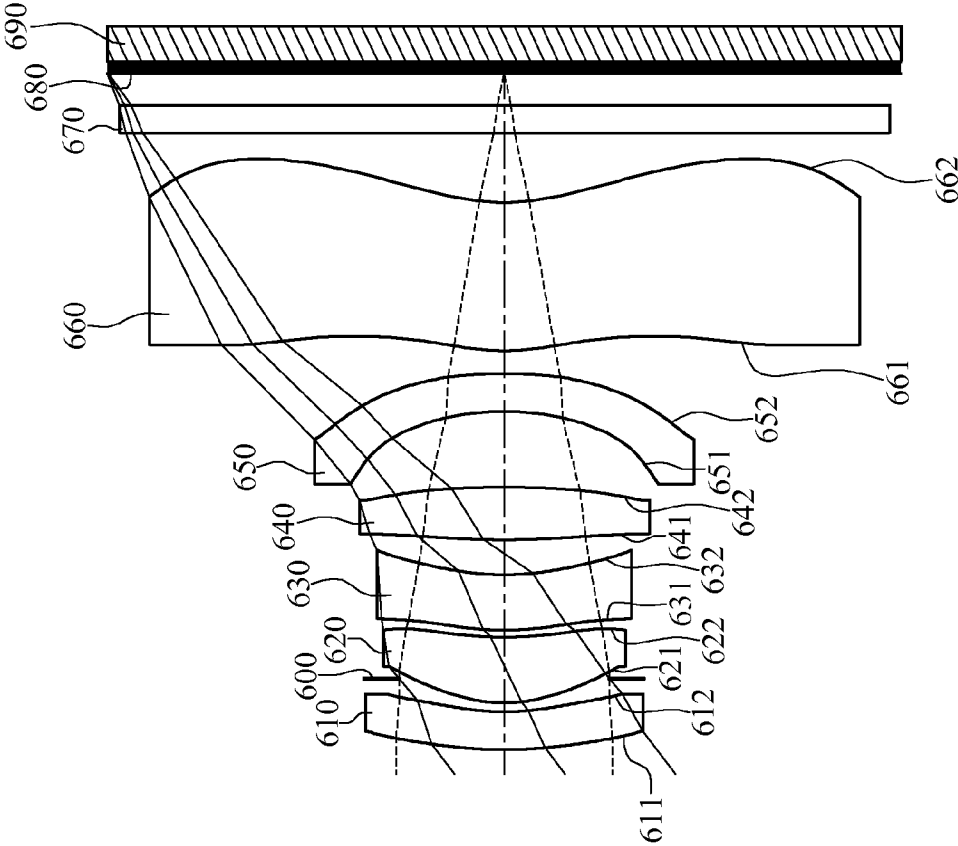
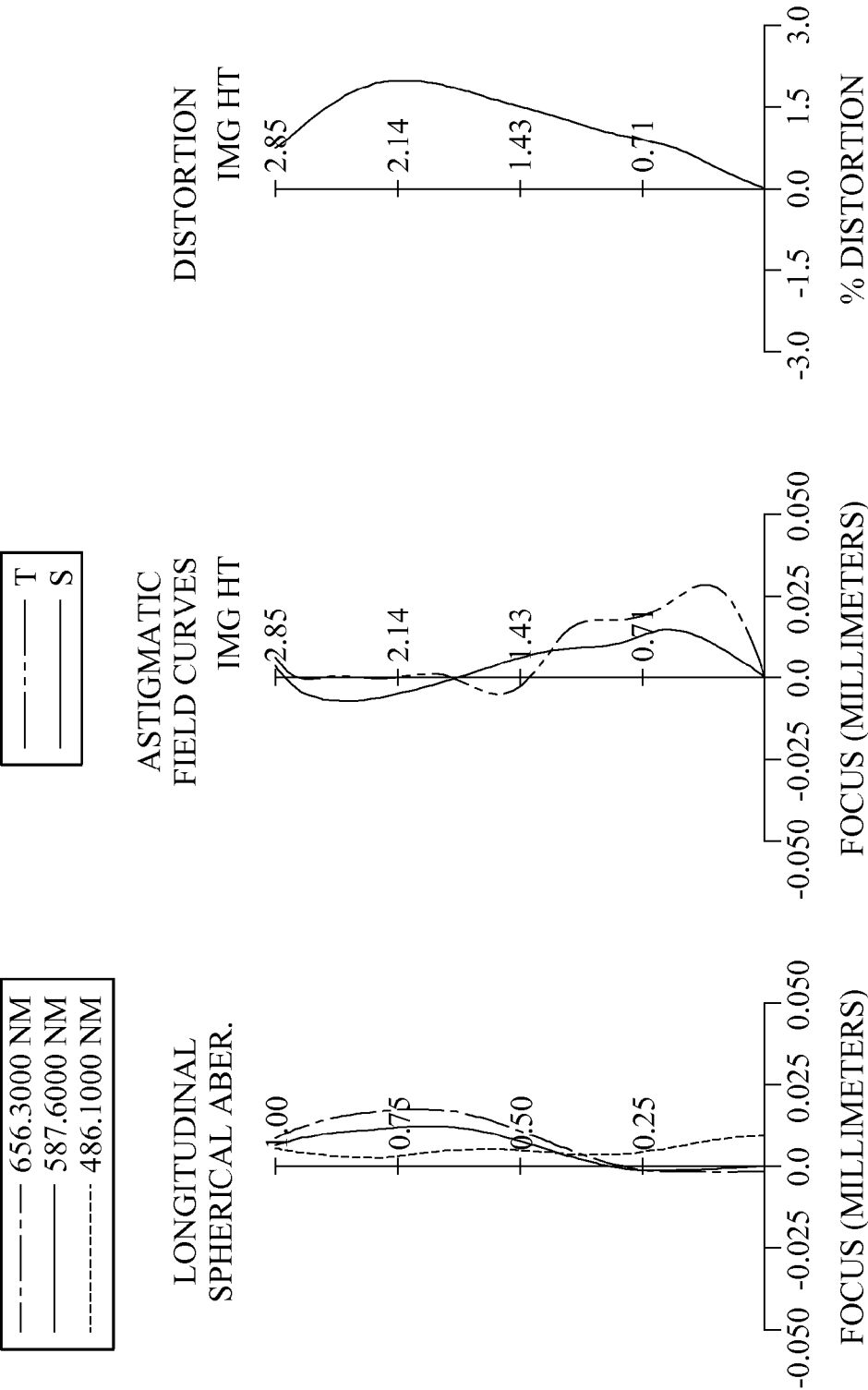


FIG.11



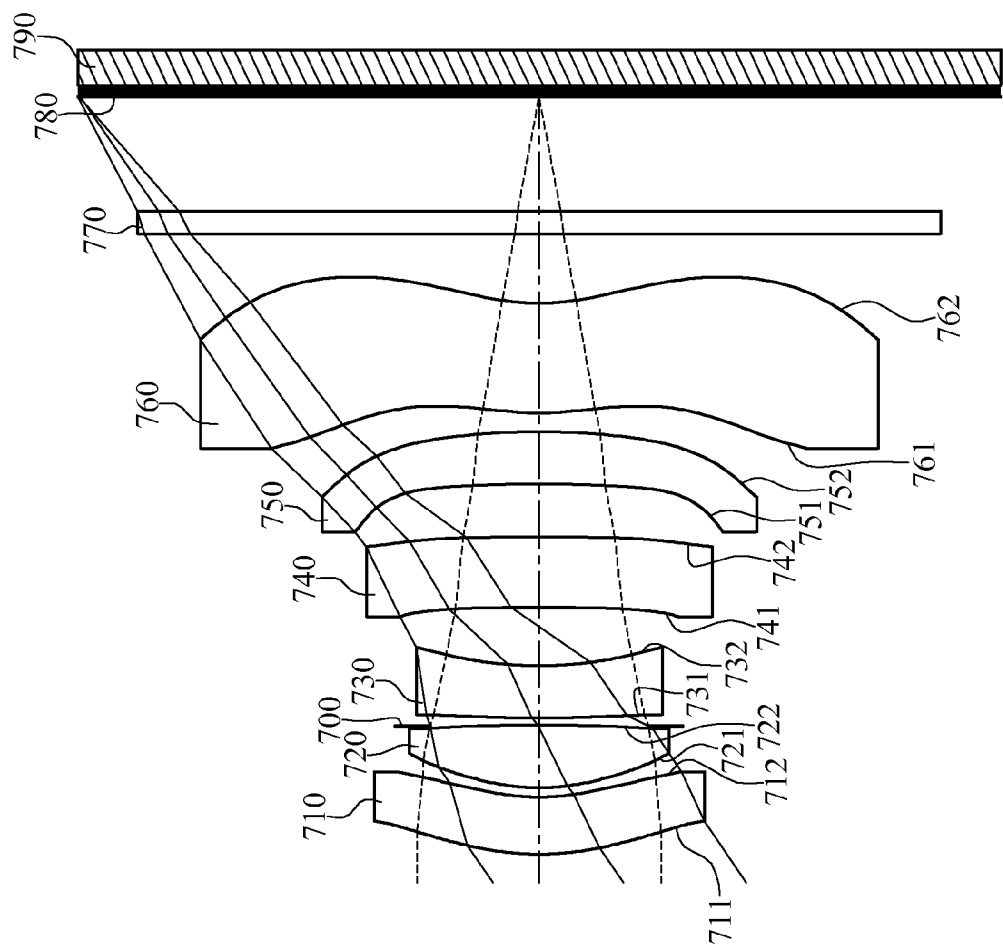


FIG.13

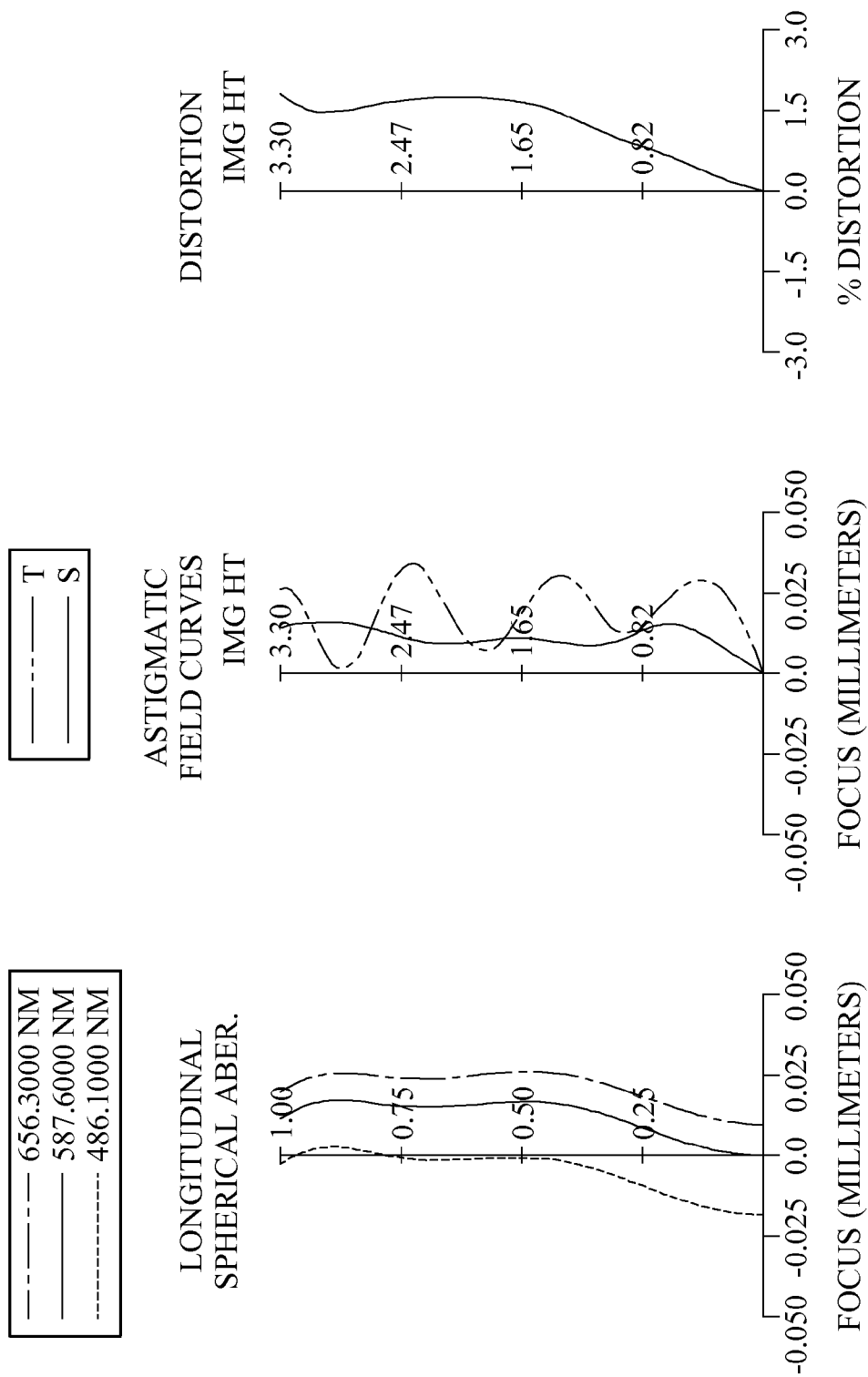


FIG. 14

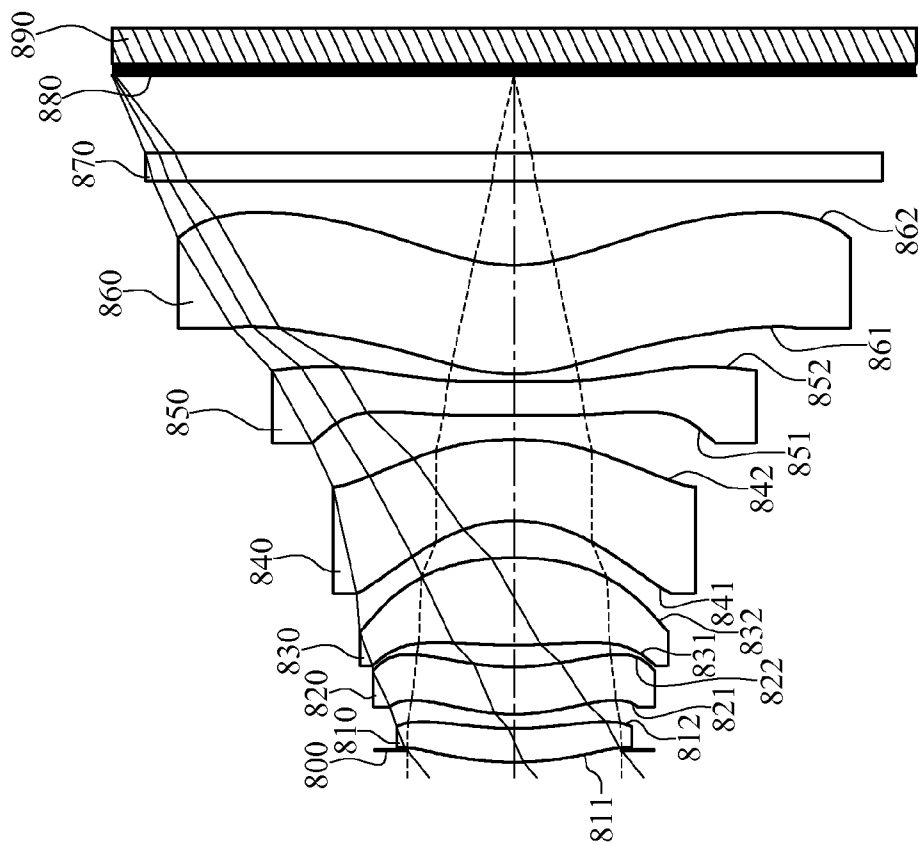


FIG.15

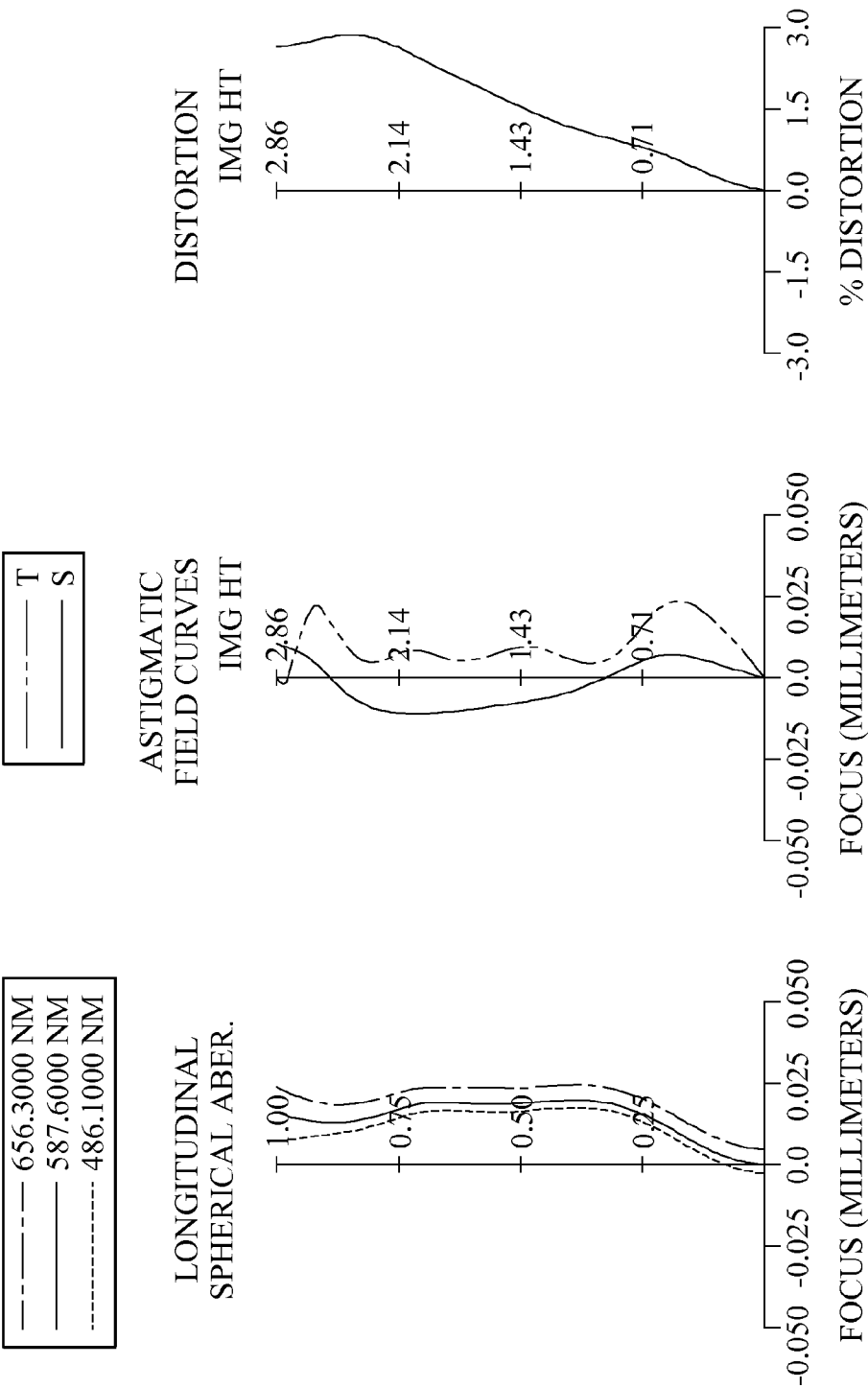


FIG.16

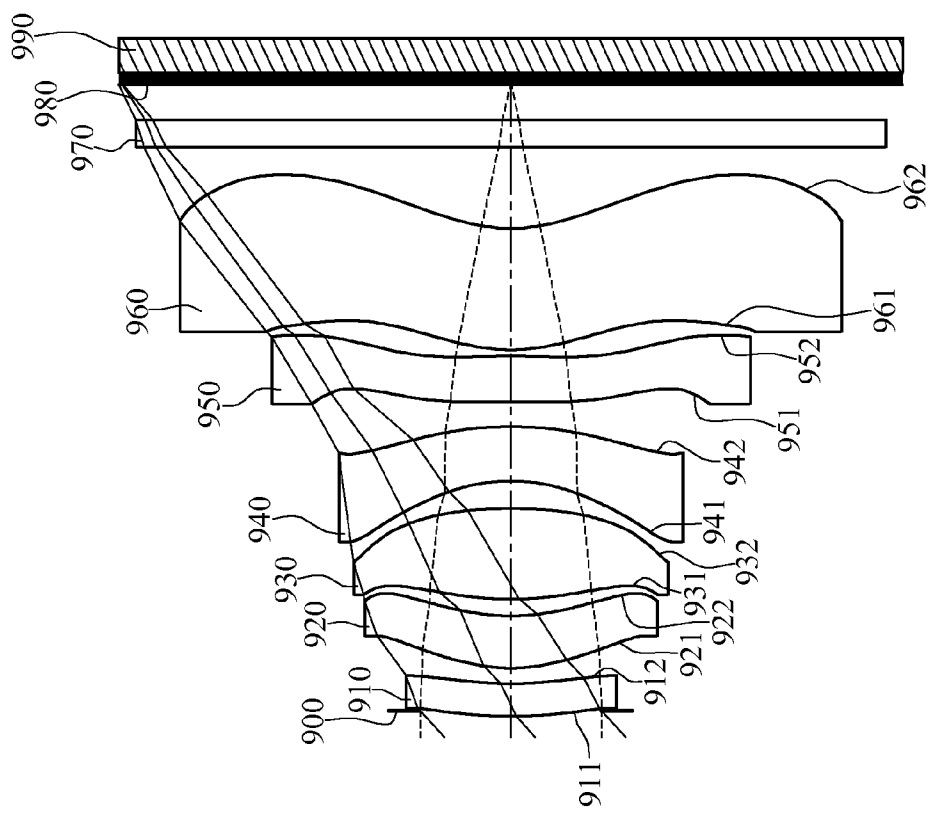


FIG.17

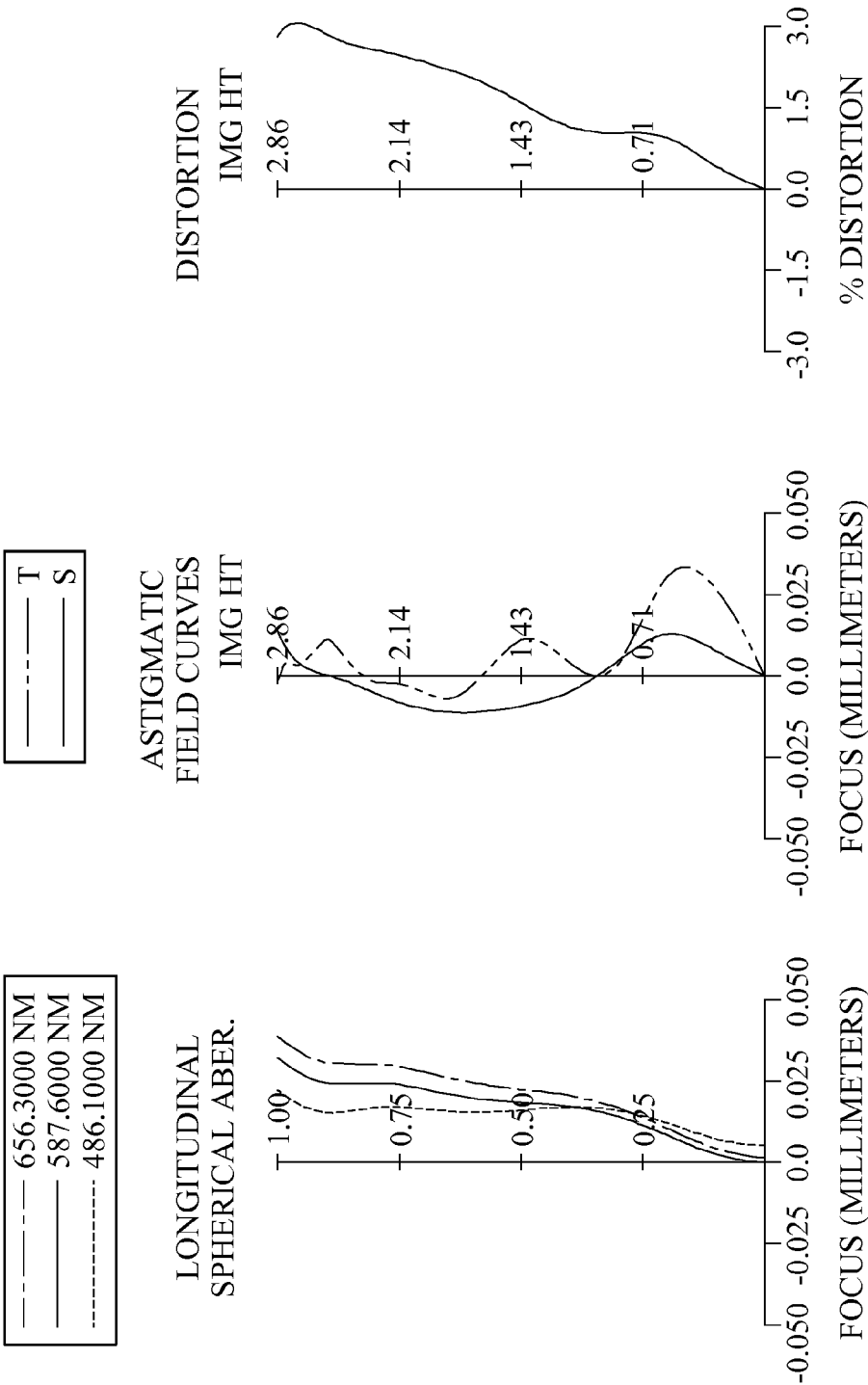


FIG.18

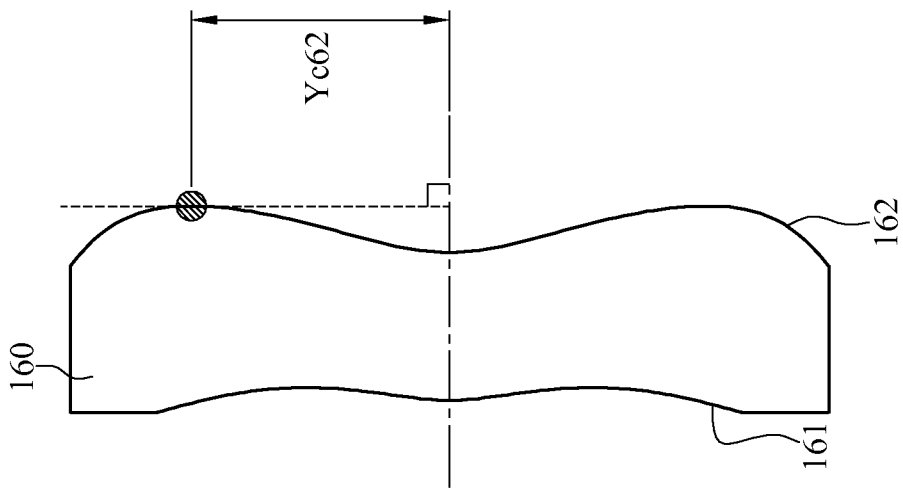


FIG.19

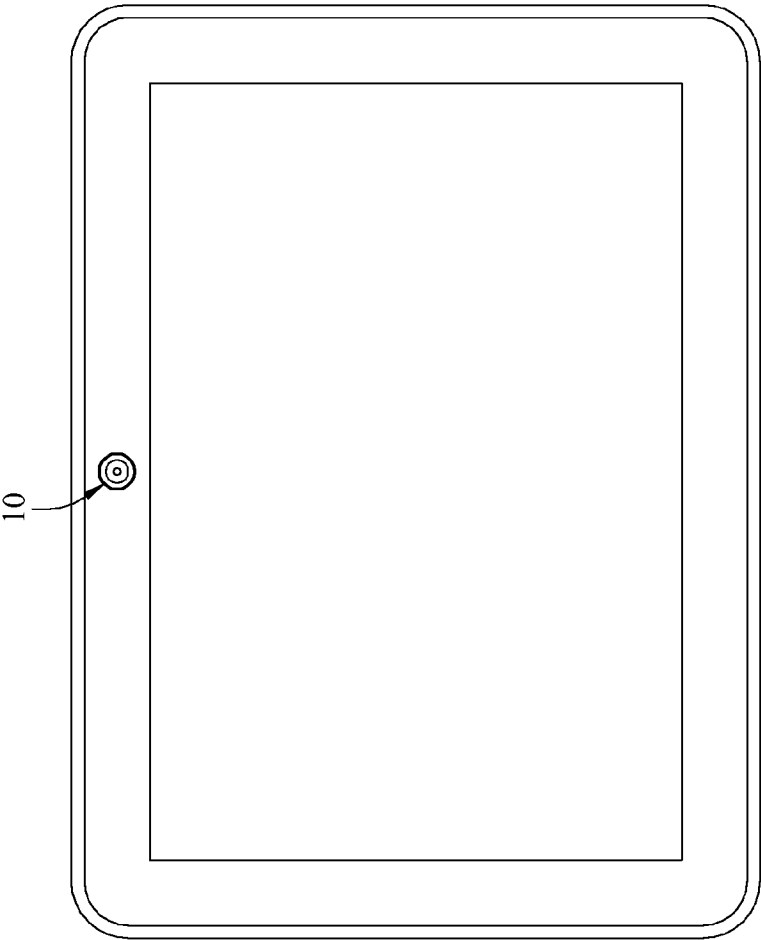


FIG. 21

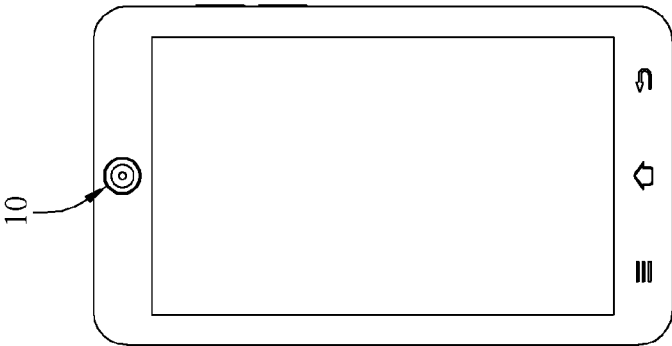


FIG. 20

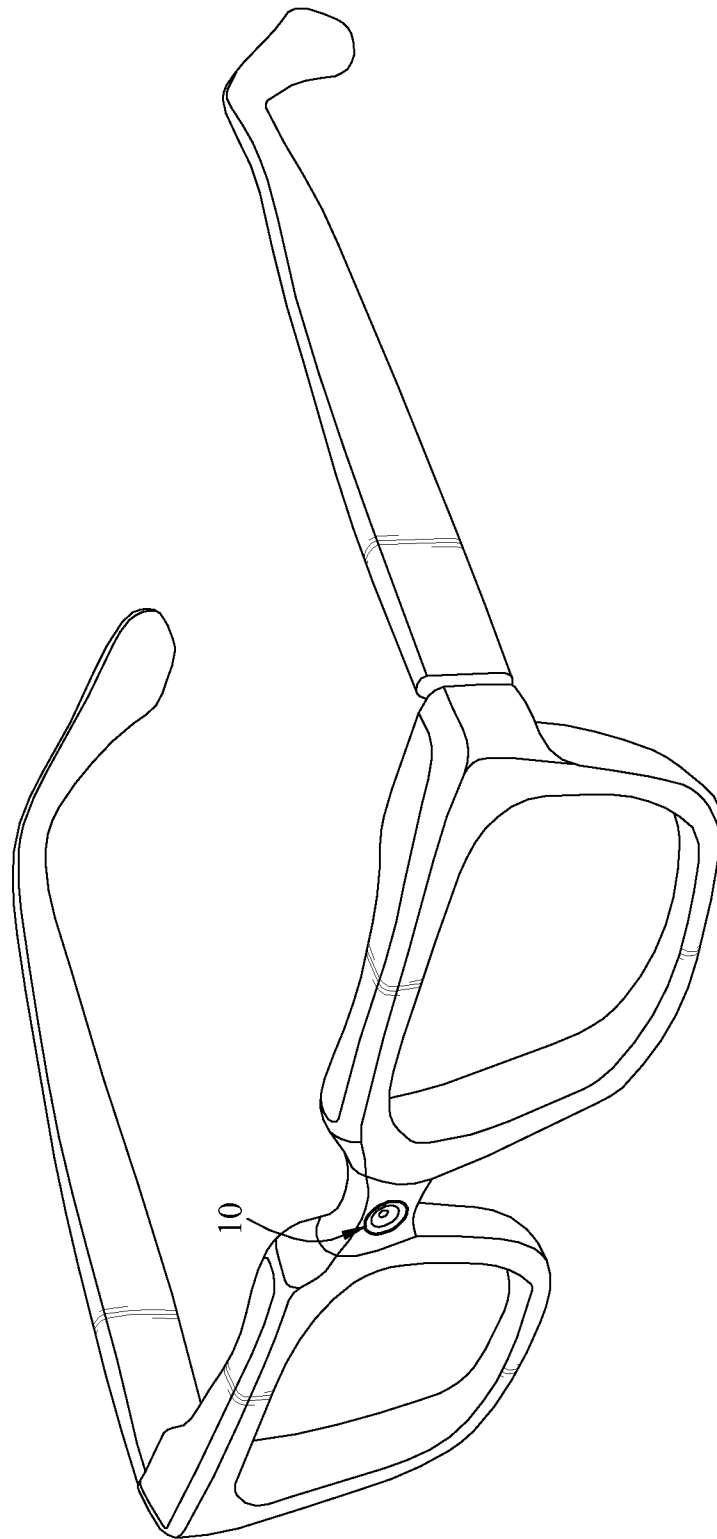


FIG. 22

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IMAGE CAPTURING LENS ASSEMBLY, IMAGE CAPTURING DEVICE AND MOBILE TERMINAL

RELATED APPLICATIONS

This application claims priority to Taiwan Application Serial Number 103102649, filed Jan. 24, 2014, which is incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

The present disclosure relates to an image capturing lens assembly, an image capturing device and a mobile terminal. More particularly, the present disclosure relates to a compact image capturing lens assembly and an image capturing device applicable to a mobile terminal.

2. Description of Related Art

In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

A conventional optical system employed in a portable electronic product mainly adopts a five-element lens structure. Due to the popularity of mobile terminals with high-end specifications, such as smart phones, tablet personal computers and wearable apparatus, the requirements for high resolution and image quality of present compact optical systems increase significantly. However, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

Other conventional compact optical systems with six-element lens structure enhance image quality and resolution. However, the field of view is not large enough to capture big image scene and the back focal length is not reduced; therefore, the total track length of the optical system is too long. It is also thereby not favorable for keeping the optical system compact.

SUMMARY

According to one aspect of the present disclosure, an image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element with negative refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element has refractive power. The fifth lens element has negative refractive power, wherein both of an object-side surface and an image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the sixth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the sixth lens element has

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at least one inflection point. The image capturing lens assembly has a total of six lens elements with refractive power. When a focal length of the image capturing lens assembly is f , a focal length of the first lens element is f_1 , a focal length of the second lens element is f_2 , and a focal length of the fifth lens element is f_5 , the following conditions are satisfied:

$$-1.0 < f_2/f_1 < 0; \text{ and}$$

$$-1.5 < f/f_5 < 0.$$

According to another aspect of the present disclosure, an image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The first lens element has negative refractive power. The second lens element has positive refractive power. The third lens element has refractive power. The fourth lens element has refractive power, wherein both of an object-side surface and an image-side surface of the fourth lens element are aspheric. The fifth lens element has refractive power, wherein both of an object-side surface and an image-side surface of the fifth lens element are aspheric. The sixth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the sixth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the sixth lens element has at least one inflection point. The image capturing lens assembly has a total of six lens elements with refractive power. When a focal length of the image capturing lens assembly is f , a focal length of the first lens element is f_1 , a focal length of the second lens element is f_2 , a focal length of the fifth lens element is f_5 , and an axial distance between an object-side surface of the first lens element and an image plane is TL , and an entrance pupil diameter of the image capturing lens assembly is EPD , the following conditions are satisfied:

$$-3.0 < f_2/f_1 < 0;$$

$$-1.5 < f/f_5 < 0.4; \text{ and}$$

$$TL/EPD < 5.0.$$

According to still another aspect of the present disclosure, an image capturing device includes the image capturing lens assembly according to the aforementioned aspect and an image sensor, wherein the image sensor is located on the image side of the image capturing lens assembly.

According to yet another aspect of the present disclosure, a mobile terminal includes the image capturing device according to the foregoing aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure;

FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment;

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure;

FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment;

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FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure;

FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment;

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure;

FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment;

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure;

FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment;

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure;

FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment;

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure;

FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment;

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure;

FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment;

FIG. 17 is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure;

FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment;

FIG. 19 shows Yc62 according to the image-side surface of the sixth lens element of the FIG. 1;

FIG. 20 shows a mobile terminal according to the 1st embodiment of the present disclosure;

FIG. 21 shows a mobile terminal according to the 2nd embodiment of the present disclosure; and

FIG. 22 shows a mobile terminal according to the 3rd embodiment of the present disclosure.

DETAILED DESCRIPTION

An image capturing lens assembly includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element, a fifth lens element, and a sixth lens element. The image capturing lens assembly has a total of six lens elements with refractive power.

The first lens element has negative refractive power, so that it is favorable for enlarging the field of view so as to capture more of the image scene. The first lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, so that the refractive variance of the incident light will be relatively minimized by reducing the angle of the incident light. It is also favorable for correcting the astigmatism of the image capturing lens assembly.

The second lens element has positive refractive power, so that it is favorable for having light convergence ability. The second lens element can have an object-side surface being convex in a paraxial region thereof; therefore, it is favorable for reinforcing the light convergence ability of the second lens element.

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The third lens element can have negative refractive power, so that it is favorable for correcting the aberration of the image capturing lens assembly. The third lens element can have an image-side surface being concave in a paraxial region thereof, so that it is favorable for the principal point of the image capturing lens assembly being positioned away from an image plane so as to reduce the total track length thereof and maintain a compact size thereof.

The fourth lens element can have positive refractive power, so that it is favorable for reducing the photosensitivity of the image capturing lens assembly.

The fifth lens element can have negative refractive power, so that it is favorable for correcting the Petzval Sum of the image capturing lens assembly so as to improve the flatness of the image plane. The fifth lens element can have an object-side surface being concave in a paraxial region thereof, so that it is favorable for properly adjusting the negative refractive power of the fifth lens element.

The sixth lens element with refractive power can have an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, so that the principal point of the image capturing lens assembly can be positioned away from the image plane so as to reduce the total track length thereof. Furthermore, at least one of the object-side surface and the image-side surface of the sixth lens element can have at least one inflection point. Therefore, it is favorable for reducing the incident angle of the off-axis on the image plane so as to correct the aberration of the off-axis.

When a focal length of the first lens element is f_1 , and a focal length of the second lens element is f_2 , the following condition is satisfied: $-3.0 < f_2/f_1 < 0$. Therefore, it is favorable for reducing the spherical aberration. Preferably, the following condition is satisfied: $-1.0 < f_2/f_1 < 0$. More preferably, the following condition is satisfied: $-0.6 < f_2/f_1 < 0$.

When a focal length of the image capturing lens assembly is f , and a focal length of the fifth lens element is f_5 , the following condition is satisfied: $-1.5 < f/f_5 < 0.4$. Therefore, it is favorable for correcting the aberration. Preferably, the following condition is satisfied: $-1.5 < f/f_5 < 0$. More preferably, the following condition is satisfied: $-0.85 < f/f_5 < 0$.

When an axial distance between the object-side surface of the first lens element and the image plane is TL , and an entrance pupil diameter of the image capturing lens assembly is EPD , the following condition is satisfied: $TL/EPD < 5.0$. Therefore, it is favorable for increasing the light exposure of the image capturing lens assembly and maintaining a compact size thereof. Preferably, the following condition is satisfied: $TL/EPD < 4.0$.

In the present disclosure, the aforementioned image capturing lens assembly can further include a stop, such as an aperture stop, wherein an axial distance between the stop and the image plane is SL , and an axial distance between the object-side surface of the first lens element and the image plane is TL , the following condition is satisfied: $0.80 < SL/TL < 1.10$. Therefore, it is favorable for reducing the total track length of the image capturing lens assembly so as to maintain a compact size thereof.

When the focal length of the image capturing lens assembly is f , and a focal length of the sixth lens element is f_6 , the following condition is satisfied: $-0.5 < f/f_6 < 0.5$. Therefore, it is favorable for the principal point of the image capturing lens assembly being positioned away from the image plane by adjusting the back focal length of the image capturing lens assembly so as to reduce the total track length thereof.

When a central thickness of the fourth lens element is CT_4 , and a central thickness of the sixth lens element is CT_6 , the

following condition is satisfied: $1.0 < CT6/CT4 < 4.0$. Therefore, the central thicknesses of the fourth and the sixth lens elements will be more proper. Therefore, it is favorable for the assembly and space arrangement of the image capturing lens assembly.

When the axial distance between the object-side surface of the first lens element and the image plane is TL, and a maximum image height of the image capturing lens assembly (half of a diagonal length of an effective photosensitive area of an image sensor) is ImgH, the following condition is satisfied: $TL/ImgH < 1.90$. Therefore, it is favorable for maintaining a compact size thereof so as to be applied to portable mobile terminals.

In the present disclosure, of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element of the image capturing lens assembly, there is a space in a paraxial region between every pair of lens elements that are adjacent to each other. Therefore, it is favorable for assembling the aspheric lens elements and reducing the assembly difficulties.

When a curvature radius of the object-side surface of the sixth lens element is R11, and a curvature radius of the image-side surface of the sixth lens element is R12, the following condition is satisfied: $-0.2 < (R11 - R12)/(R11 + R12) < 0.4$. Therefore, the principal point of the image capturing lens assembly can be positioned away from the image plane, so that it is favorable for reducing the total track length to maintain a compact size thereof.

When an axial distance between the fourth lens element and the fifth lens element is T45, and an axial distance between the fifth lens element and the sixth lens element is T56, the following condition is satisfied: $0 < T56/T45 < 0.90$. Therefore, it is favorable for assembling the lens elements so as to increase manufacturing yield rate.

When an Abbe number of the second lens element is V2, the following condition is satisfied: $40 < V2 < 70$. Therefore, it is favorable for correcting the chromatic aberration of the image capturing lens assembly.

When an axial distance between the second lens element and the third lens element is T23, and an axial distance between the third lens element and the fourth lens element is T34, the following condition is satisfied: $0 < T23/T34 < 0.50$. Therefore, it is favorable for assembling the lens elements by properly adjusting the axial distances between the aforementioned lens elements so as to reduce the total track length. It is thereby favorable for keeping the size compact to be applied to portable mobile terminals.

When a curvature radius of an object-side surface of the fourth lens element is R7, and a curvature radius of an image-side surface of the fourth lens element is R8, the following condition is satisfied: $-0.5 < (R7 + R8)/(R7 - R8) < 0.5$. Therefore, it is favorable for correcting the astigmatism.

When a vertical distance between a non-axial critical point on the image-side surface of the sixth lens element and an optical axis is Yc62, and the axial distance between the object-side surface of the first lens element and the image plane is TL, the following condition is satisfied: $0.1 < Yc62/TL < 0.7$. Therefore, it is favorable for effectively correcting the aberration of the off-axis and reducing the total track length of the image capturing lens assembly so as to keep a compact size thereof. Particularly, FIG. 19 shows Yc62 according to the image-side surface of the sixth lens element of the FIG. 1. The critical point on the image-side surface of the sixth lens element is a non-axial point of the lens surface where its tangent is perpendicular to an optical axis.

According to the image capturing lens assembly of the present disclosure, the lens elements thereof can be made of glass or plastic material. When the lens elements are made of glass material, the distribution of the refractive power of the image capturing lens assembly may be more flexible to design. When the lens elements are made of plastic material, the manufacturing cost can be effectively reduced. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the image capturing lens assembly can also be reduced.

According to the image capturing lens assembly of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface where light rays travel away from the optical axis. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof.

According to the image capturing lens assembly of the present disclosure, a critical point is a non-axial point of the lens surface where its tangent is perpendicular to an optical axis.

According to the image capturing lens assembly of the present disclosure, the image capturing lens assembly can include at least one stop, such as an aperture stop, a glare stop or a field stop. Said glare stop or said field stop is for eliminating the stray light and thereby improving the image resolution thereof.

According to the image capturing lens assembly of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the image capturing lens assembly and the image plane and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image plane is favorable for enlarging the field of view of the image capturing lens assembly and thereby provides a wider field of view for the same.

According to the image capturing lens assembly of the present disclosure, the image capturing lens assembly can be optionally applied to moving focus optical systems. Furthermore, the image capturing lens assembly is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, wearable devices and other mobile terminals.

According to the present disclosure, an image capturing device is provided. The image capturing device includes the aforementioned image capturing lens assembly according to the present disclosure and an image sensor, wherein the image sensor is disposed on or near the image plane of the image capturing lens assembly. Preferably, the image capturing device can further include a barrel member, a holding member or a combination thereof.

In FIG. 20, FIG. 21 and FIG. 22, an image capturing device 10 may be installed in, but not limited to, a mobile terminal, including a smart phone (FIG. 20), a tablet personal computer (FIG. 21) or a wearable device (FIG. 22). The three exemplary figures of different kinds of mobile terminal are only exem-

plary for showing the image capturing device of present disclosure installing in a mobile terminal and is not limited thereto. In some embodiments, the mobile terminal can further include, but not limited to, a display, a control unit, a random access memory unit (RAM), a read only memory unit (ROM) or a combination thereof.

According to the above description of the present disclosure, the following specific embodiments are provided for further explanation.

1st Embodiment

FIG. 1 is a schematic view of an image capturing device according to the 1st embodiment of the present disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 1st embodiment.

In FIG. 1, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 190. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 110, an aperture stop 100, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, a sixth lens element 160, an IR-cut filter 170 and an image plane 180, wherein the image capturing lens assembly has a total of six lens elements (110-160) with refractive power.

The first lens element 110 with negative refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 110 is made of plastic material.

The second lens element 120 with positive refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being concave in a paraxial region thereof, which are both aspheric, and the second lens element 120 is made of plastic material.

The third lens element 130 with negative refractive power has an object-side surface 131 being convex in a paraxial region thereof and an image-side surface 132 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 130 is made of plastic material.

The fourth lens element 140 with positive refractive power has an object-side surface 141 being convex in a paraxial region thereof and an image-side surface 142 being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element 140 is made of plastic material.

The fifth lens element 150 with negative refractive power has an object-side surface 151 being concave in a paraxial region thereof and an image-side surface 152 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 150 is made of plastic material.

The sixth lens element 160 with positive refractive power has an object-side surface 161 being convex in a paraxial region thereof and an image-side surface 162 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 160 is made of plastic material. Specifically, both of the object-side surface 161 and the image-side surface 162 of the sixth lens element 160 have at least one inflection point.

The IR-cut filter 170 is made of glass and located between the sixth lens element 160 and the image plane 180, and will not affect the focal length of the image capturing lens assembly. The image sensor 190 is disposed on or near the image plane 180 of the image capturing lens assembly.

The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

where,

X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

Y is the vertical distance from the point on the aspheric surface to the optical axis;

R is the curvature radius;

k is the conic coefficient; and

Ai is the i-th aspheric coefficient.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a focal length of the image capturing lens assembly is f, an f-number of the image capturing lens assembly is Fno, and half of a maximal field of view of the image capturing lens assembly is HFOV, these parameters have the following values: f=3.73 mm; Fno=2.20; and HFOV=37.2 degrees.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when an Abbe number of the second lens element 120 is V2, the following condition is satisfied: V2=55.9.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, a central thickness of the fourth lens element 140 is CT4, and a central thickness of the sixth lens element 160 is CT6, the following condition is satisfied: CT6/CT4=2.82.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when an axial distance between the second lens element 120 and the third lens element 130 is T23, and an axial distance between the third lens element 130 and the fourth lens element 140 is T34, the following condition is satisfied: T23/T34=0.18.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when an axial distance between the fourth lens element 140 and the fifth lens element 150 is T45, and an axial distance between the fifth lens element 150 and the sixth lens element 160 is T56, the following condition is satisfied: T56/T45=0.32.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a curvature radius of the object-side surface 141 of the fourth lens element 140 is R7, and a curvature radius of the image-side surface 142 of the fourth lens element 140 is R8, the following condition is satisfied: (R7+R8)/(R7-R8)=0.36.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a curvature radius of the object-side surface 161 of the sixth lens element 160 is R11, and a curvature radius of the image-side surface 162 of the sixth lens element 160 is R12, the following condition is satisfied: (R11-R12)/(R11+R12)=0.05.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when a focal length of the first lens element 110 is f1, and a focal length of the second lens element 120 is f2, the following condition is satisfied: f2/f1=-0.24.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the focal

length of the image capturing lens assembly is f , and a focal length of the fifth lens element **150** is f_5 , the following condition is satisfied: $f/f_5 = -0.53$.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the focal length of the image capturing lens assembly is f , and a focal length of the sixth lens element **160** is f_6 , the following condition is satisfied: $f/f_6 = 0.17$.

FIG. **19** shows Yc62 according to the image-side surface **162** of the sixth lens element **160** of the FIG. **1**. In FIG. **19**, when a vertical distance between a non-axial critical point on the image-side surface **162** of the sixth lens element **160** and an optical axis is Yc62, and an axial distance between the object-side surface **111** of the first lens element **110** and the image plane **180** is TL, the following condition is satisfied: Yc62/TL=0.37.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when an axial distance between the aperture stop **100** and the image plane **180** is SL, and the axial distance between the object-side

surface **111** of the first lens element **110** and the image plane **180** is TL, and the following condition is satisfied: SL/TL=0.88.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the axial distance between the object-side surface **111** of the first lens element **110** and the image plane **180** is TL, and an entrance pupil diameter of the image capturing lens assembly is EPD, the following condition is satisfied: TL/EPD=2.78.

In the image capturing lens assembly of the image capturing device according to the 1st embodiment, when the axial distance between the object-side surface **111** of the first lens element **110** and the image plane **180** is TL, and a maximum image height of the image capturing lens assembly (half of a diagonal length of an effective photosensitive area of an image sensor **190**) is ImgH, the following condition is satisfied: TL/ImgH=1.65.

The detailed optical data of the 1st embodiment are shown in Table 1 and the aspheric surface data are shown in Table 2 below.

TABLE 1

1st Embodiment								
$f = 3.73 \text{ mm}$, $Fno = 2.20$, $HFOV = 37.2 \text{ deg}$.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	1.887 ASP	0.343	Plastic	1.530	55.8	-10.44	
2		1.318 ASP	0.216					
3	Ape. Stop	Plano	-0.177					
4	Lens 2	1.088 ASP	0.443	Plastic	1.544	55.9	2.48	
5		4.854 ASP	0.050					
6	Lens 3	5.099 ASP	0.240	Plastic	1.640	23.3	-7.54	
7		2.434 ASP	0.272					
8	Lens 4	11.525 ASP	0.354	Plastic	1.544	55.9	6.81	
9		-5.404 ASP	0.541					
10	Lens 5	-1.741 ASP	0.325	Plastic	1.640	23.3	-6.97	
11		-3.063 ASP	0.173					
12	Lens 6	1.608 ASP	1.000	Plastic	1.544	55.9	21.83	
13		1.453 ASP	0.500					
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—	
15		Plano	0.223					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	-4.2721E-01	-7.1390E+00	-4.3409E+00	4.6581E+00	-1.8839E+01	-1.1488E+01
A4 =	-5.4031E-02	-1.7315E-01	-1.3845E-02	-2.9771E-02	-2.1825E-01	-9.9796E-02
A6 =	5.9177E-02	2.8878E-02	-5.9499E-03	1.2998E-01	3.5191E-01	2.9299E-01
A8 =	-1.0589E-01	1.2407E-01	-7.6694E-02	-7.9203E-01	-3.3747E-01	-5.1716E-01
A10 =	1.4137E-01	-2.4389E-02	2.0129E-02	1.5469E-01	-4.4605E-01	1.1826E+00
A12 =	-9.3821E-02	-5.1438E-02	9.5847E-02	1.1399E+00	1.4191E+00	-1.3868E+00
A14 =	3.4320E-02	3.6685E-02	-1.6624E-01	-7.9308E-01	-7.8293E-01	6.3741E-01
Surface #						
	8	9	10	11	12	13
k =	-4.2103E+00	-9.8252E-01	-2.3812E-01	3.0534E+00	-2.0000E+01	-5.8181E+00
A4 =	-3.2154E-02	2.3481E-02	1.1447E-01	-3.0654E-01	-1.1913E-01	-5.3305E-02
A6 =	-1.7932E-01	-2.2218E-01	-4.2201E-01	7.6871E-01	5.7962E-02	1.8193E-02
A8 =	4.0532E-01	4.6796E-01	1.1680E+00	-1.2423E+00	-1.4355E-02	-4.6816E-03
A10 =	-5.4477E-01	-6.9547E-01	-2.6388E+00	1.2329E+00	8.3195E-04	7.9393E-04
A12 =	8.5550E-03	4.9832E-01	3.3900E+00	-7.3118E-01	3.3814E-04	-8.9826E-05

TABLE 2-continued

Aspheric Coefficients					
A14 =	6.6292E-01	-1.1544E-01	-2.3165E+00	2.3069E-01	-6.3671E-05
A16 =	-4.2904E-01		6.3539E-01	-2.8966E-02	3.1667E-06

In Table 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-16 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In Table 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as Table 1 and Table 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

FIG. 3 is a schematic view of an image capturing device according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 2nd embodiment.

In FIG. 3, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 290. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 210, an aperture stop 200, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, a sixth lens element 260, an IR-cut filter 270 and an image plane 280, wherein the image capturing lens assembly has a total of six lens elements (210-260) with refractive power.

The first lens element 210 with negative refractive power has an object-side surface 211 being convex in a paraxial region thereof and an image-side surface 212 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 210 is made of plastic material.

The second lens element 220 with positive refractive power has an object-side surface 221 being convex in a paraxial region thereof and an image-side surface 222 being concave in a paraxial region thereof, which are both aspheric, and the second lens element 220 is made of plastic material.

The third lens element 230 with negative refractive power has an object-side surface 231 being convex in a paraxial region thereof and an image-side surface 232 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 230 is made of plastic material.

The fourth lens element 240 with positive refractive power has an object-side surface 241 being convex in a paraxial region thereof and an image-side surface 242 being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element 240 is made of plastic material.

The fifth lens element 250 with negative refractive power has an object-side surface 251 being concave in a paraxial region thereof and an image-side surface 252 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 250 is made of plastic material.

The sixth lens element 260 with negative refractive power has an object-side surface 261 being convex in a paraxial region thereof and an image-side surface 262 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 260 is made of plastic material. Specifically, both of the object-side surface 261 and the image-side surface 262 of the sixth lens element 260 have at least one inflection point.

The IR-cut filter 270 is made of glass and located between the sixth lens element 260 and the image plane 280, and will not affect the focal length of the image capturing lens assembly. The image sensor 290 is disposed on or near the image plane 280 of the image capturing lens assembly.

The detailed optical data of the 2nd embodiment are shown in Table 3 and the aspheric surface data are shown in Table 4 below.

TABLE 3

2nd Embodiment								
f = 3.63 mm, Fno = 2.00, HFOV = 37.7 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	2.148 ASP	0.287	Plastic	1.544	55.9	-15.84	
2		1.639 ASP	0.340					
3	Ape. Stop	Plano	-0.245					
4	Lens 2	1.192 ASP	0.468	Plastic	1.544	55.9	2.70	
5		5.425 ASP	0.050					
6	Lens 3	4.115 ASP	0.240	Plastic	1.650	21.4	-8.09	
7		2.255 ASP	0.274					
8	Lens 4	7.921 ASP	0.319	Plastic	1.544	55.9	9.63	
9		-15.241 ASP	0.460					
10	Lens 5	-3.291 ASP	0.358	Plastic	1.650	21.4	-14.13	
11		-5.351 ASP	0.219					
12	Lens 6	1.687 ASP	0.875	Plastic	1.535	55.7	-307.39	
13		1.368 ASP	0.500					
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—	
15		Plano	0.255					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	-5.6980E+00	-7.8065E+00	-3.5250E+00	5.0000E+00	5.0000E+00	-7.3303E+00
A4 =	-5.4730E-02	-1.7616E-01	5.0297E-02	4.4225E-02	-1.6348E-01	-9.3348E-02
A6 =	5.1021E-02	-1.5742E-02	-3.8843E-02	1.4640E-01	3.1091E-01	3.0035E-01
A8 =	-1.2823E-01	1.0233E-01	-5.4938E-02	-7.0729E-01	-4.1720E-01	-6.1920E-01
A10 =	1.4424E-01	-5.5386E-03	1.3055E-01	9.0580E-04	-3.9810E-01	1.1232E+00
A12 =	-7.3729E-02	-5.2000E-02	-1.0549E-01	1.0475E+00	1.2511E+00	-1.0794E+00
A14 =	1.5148E-02	2.2203E-02	7.3346E-03	-6.0565E-01	-6.2335E-01	4.2552E-01
Surface #						
	8	9	10	11	12	13
k =	-9.1403E+00	-3.0046E+00	2.4239E+00	5.0000E+00	-2.0000E+01	-5.8540E+00
A4 =	-4.9791E-02	9.5233E-03	6.7444E-02	-2.7326E-01	-1.2770E-01	-6.1824E-02
A6 =	-1.3533E-01	-2.4610E-01	-2.9814E-01	7.2487E-01	5.5233E-02	2.1945E-02
A8 =	3.0860E-01	4.7776E-01	9.5903E-01	-1.2251E+00	-1.3470E-02	-6.2072E-03
A10 =	-4.0466E-01	-6.7492E-01	-2.4990E+00	1.2145E+00	1.1303E-03	1.0966E-03
A12 =	2.3961E-02	4.9211E-01	3.3620E+00	-7.2531E-01	3.0150E-04	-1.1725E-04
A14 =	4.8765E-01	-1.1429E-01	-2.3376E+00	2.3476E-01	-8.0164E-05	5.4524E-06
A16 =	-3.2062E-01		6.4736E-01	-3.0836E-02	5.4761E-06	

In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 3 and Table 4 as the following values and satisfy the following conditions:

2nd Embodiment			
f [mm]	3.63	(R11 - R12)/(R11 + R12)	0.10
Fno	2.00	f2/f1	-0.17
HFOV [deg.]	37.7	f/f5	-0.26
V2	55.9	f/f6	-0.01
CT6/CT4	2.74	Yc62/TL	0.35
T23/T34	0.18	SL/TL	0.86
T56/T45	0.48	TL/EPD	2.54
(R7 + R8)/(R7 - R8)	-0.32	TL/lmgH	1.61

3rd Embodiment

FIG. 5 is a schematic view of an image capturing device according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 3rd embodiment.

In FIG. 5, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 390. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 310, an aperture stop 300, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, a sixth lens element 360, an IR-cut filter 370 and an image plane 380, wherein the image capturing lens assembly has a total of six lens elements (310-360) with refractive power.

The first lens element 310 with negative refractive power has an object-side surface 311 being convex in a paraxial

region thereof and an image-side surface 312 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 310 is made of plastic material.

The second lens element 320 with positive refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being concave in a paraxial region thereof, which are both aspheric, and the second lens element 320 is made of plastic material.

The third lens element 330 with negative refractive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 330 is made of plastic material.

The fourth lens element 340 with positive refractive power has an object-side surface 341 being convex in a paraxial region thereof and an image-side surface 342 being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element 340 is made of plastic material.

The fifth lens element 350 with negative refractive power has an object-side surface 351 being concave in a paraxial region thereof and an image-side surface 352 being concave in a paraxial region thereof, which are both aspheric, and the fifth lens element 350 is made of plastic material.

The sixth lens element 360 with positive refractive power has an object-side surface 361 being convex in a paraxial region thereof and an image-side surface 362 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 360 is made of plastic material. Specifically, both of the object-side surface 361 and the image-side surface 362 of the sixth lens element 360 have at least one inflection point.

The IR-cut filter 370 is made of glass and located between the sixth lens element 360 and the image plane 380, and will not affect the focal length of the image capturing lens assembly. The image sensor 390 is disposed on or near the image plane 380 of the image capturing lens assembly.

The detailed optical data of the 3rd embodiment are shown in Table 5 and the aspheric surface data are shown in Table 6 below.

TABLE 5

3rd Embodiment f = 3.27 mm, Fno = 2.10, HFOV = 42.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	2.152 ASP	0.209	Plastic	1.572	56.0	-31.82	
2		1.856 ASP	0.281					
3	Ape. Stop	Plano	-0.160					
4	Lens 2	1.452 ASP	0.409	Plastic	1.572	56.0	2.95	
5		9.347 ASP	0.050					
6	Lens 3	3.497 ASP	0.240	Plastic	1.639	23.5	-8.02	
7		2.023 ASP	0.268					
8	Lens 4	14.953 ASP	0.343	Plastic	1.572	56.0	14.65	
9		-18.904 ASP	0.346					
10	Lens 5	-19.928 ASP	0.331	Plastic	1.639	23.5	-18.38	
11		28.794 ASP	0.289					
12	Lens 6	1.298 ASP	0.753	Plastic	1.572	56.0	13.29	
13		1.235 ASP	0.500					
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—	
15		Plano	0.304					
16	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients						
Surface #						
	1	2	4	5	6	7
k =	-7.7292E+00	-9.9551E+00	-5.0357E+00	-1.3728E+01	-1.4498E+01	-2.9779E+00
A4 =	-8.1254E-02	-1.6583E-01	6.4047E-02	-5.3279E-02	-1.3309E-01	-6.7139E-02
A6 =	2.7219E-02	-1.0793E-02	-5.6911E-02	9.7348E-02	2.1723E-01	2.3577E-01
A8 =	-8.4889E-02	7.8378E-02	-2.6696E-02	-3.0086E-01	-3.2578E-01	-4.5202E-01
A10 =	9.6244E-02	-1.3394E-02	1.1972E-01	-2.7441E-02	-8.7000E-02	6.1902E-01
A12 =	-4.9565E-02	-3.6561E-02	-9.3354E-02	6.2678E-01	7.2863E-01	-5.1298E-01
A14 =	1.1378E-02	2.0470E-02	2.4259E-02	-3.5996E-01	-4.1022E-01	1.9546E-01
Surface #						
	8	9	10	11	12	13
k =	2.8389E+00	-2.0000E+01	5.0000E+00	-2.0000E+01	-7.5428E+00	-4.4514E+00
A4 =	-6.9742E-02	-5.1967E-02	1.6131E-02	-2.8769E-01	-1.3168E-01	-7.1558E-02
A6 =	4.8713E-03	-1.6778E-01	-2.0282E-01	6.1536E-01	4.7565E-02	2.2757E-02
A8 =	8.5464E-02	3.2940E-01	5.8772E-01	-8.9972E-01	-9.8489E-03	-5.3898E-03
A10 =	-1.3054E-01	-4.2136E-01	-1.4957E+00	7.7875E-01	6.7892E-04	7.6087E-04
A12 =	7.9621E-02	2.9004E-01	1.9048E+00	-4.1674E-01	1.8011E-04	-7.3262E-05
A14 =	9.1185E-02	-5.3573E-02	-1.2511E+00	1.2505E-01	-3.9509E-05	3.6035E-06
A16 =	-9.7056E-02		3.2724E-01	-1.5650E-02	2.2158E-06	

In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 5 and Table 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f [mm]	3.27	(R11 - R12)/(R11 + R12)	0.02
Fno	2.10	f2/f1	-0.09

-continued

3rd Embodiment			
HFOV [deg.]	42.0	f/f5	-0.18
V2	56.0	f/f6	0.25
CT6/CT4	2.20	Yc62/TL	0.36
T23/T34	0.19	SL/TL	0.89
T56/T45	0.84	TL/EPD	2.81
(R7 + R8)/(R7 - R8)	-0.12	TL/lmgH	1.45

4th Embodiment

FIG. 7 is a schematic view of an image capturing device according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration

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curves, astigmatic field curves and a distortion curve of the image capturing device according to the 4th embodiment.

In FIG. 7, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 490. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 400, a first lens element 410, a second lens element 420, a third lens element 430, a fourth lens element 440, a fifth lens element 450, a sixth lens element 460, an IR-cut filter 470 and an image plane 480, wherein the image capturing lens assembly has a total of six lens elements (410-460) with refractive power.

The first lens element 410 with negative refractive power has an object-side surface 411 being convex in a paraxial region thereof and an image-side surface 412 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 410 is made of plastic material.

The second lens element 420 with positive refractive power has an object-side surface 421 being convex in a paraxial region thereof and an image-side surface 422 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 420 is made of plastic material.

The third lens element 430 with negative refractive power has an object-side surface 431 being convex in a paraxial region thereof and an image-side surface 432 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 430 is made of plastic material.

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The fourth lens element 440 with positive refractive power has an object-side surface 441 being convex in a paraxial region thereof and an image-side surface 442 being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element 440 is made of plastic material.

The fifth lens element 450 with negative refractive power has an object-side surface 451 being concave in a paraxial region thereof and an image-side surface 452 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 450 is made of plastic material.

The sixth lens element 460 with negative refractive power has an object-side surface 461 being convex in a paraxial region thereof and an image-side surface 462 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 460 is made of plastic material. Specifically, both of the object-side surface 461 and the image-side surface 462 of the sixth lens element 460 have at least one inflection point.

The IR-cut filter 470 is made of glass and located between the sixth lens element 460 and the image plane 480, and will not affect the focal length of the image capturing lens assembly. The image sensor 490 is disposed on or near the image plane 480 of the image capturing lens assembly.

The detailed optical data of the 4th embodiment are shown in Table 7 and the aspheric surface data are shown in Table 8 below.

TABLE 7

4th Embodiment							
f = 4.55 mm, Fno = 1.90, HFOV = 37.6 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.185				
2	Lens 1	2.368 ASP	0.401	Plastic	1.560	56.0	-71.22
3		2.100 ASP	0.144				
4	Lens 2	1.989 ASP	0.579	Plastic	1.535	55.7	3.68
5		-163.526 ASP	0.050				
6	Lens 3	5.290 ASP	0.312	Plastic	1.639	23.5	-5.67
7		2.101 ASP	0.172				
8	Lens 4	7.684 ASP	0.712	Plastic	1.535	55.7	6.36
9		-5.915 ASP	0.580				
10	Lens 5	-15.858 ASP	0.738	Plastic	1.639	23.5	-36.93
11		-49.231 ASP	0.139				
12	Lens 6	2.148 ASP	0.850	Plastic	1.535	55.7	-15.79
13		1.476 ASP	0.700				
14	IR-cut filter	Plano	0.150	Glass	1.517	64.2	—
15		Plano	0.404				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 8

Aspheric Coefficients						
	Surface #					
	2	3	4	5	6	7
k =	-2.9886E+00	-4.8206E+00	-4.2970E+00	-2.6710E+00	-1.9984E+01	-4.6660E+00
A4 =	-2.3095E-02	-7.7276E-02	8.7409E-04	6.7928E-02	-3.2817E-02	-2.7111E-02
A6 =	-1.9804E-03	-2.8578E-02	-2.2610E-02	1.2522E-02	3.7923E-02	4.0992E-02
A8 =	-2.6898E-03	1.0075E-02	-5.2309E-03	-2.5227E-02	-2.8382E-02	-4.5463E-02
A10 =	-5.2324E-04	8.4634E-03	6.8344E-03	-7.8041E-03	-6.3503E-03	2.2046E-02
A12 =	-1.4406E-03	-7.2517E-03	1.0510E-03	8.7892E-03	1.0622E-02	-4.8462E-03
A14 =	5.7495E-04	1.6673E-03	-1.0142E-03	-1.9027E-03	-2.4150E-03	3.6709E-04

TABLE 8-continued

Aspheric Coefficients						
Surface #						
	8	9	10	11	12	13
k =	-8.0101E-01	2.8574E+00	-9.7133E+00	1.5488E+00	-1.5306E+01	-4.8589E+00
A4 =	1.8137E-02	-4.1161E-03	4.2202E-02	-4.7263E-02	-9.0507E-02	-4.6497E-02
A6 =	4.1092E-03	-2.2113E-02	-7.3603E-02	7.1289E-02	1.7675E-02	1.0020E-02
A8 =	5.5748E-03	2.6087E-02	6.4176E-02	-6.4313E-02	-2.8586E-03	-1.5213E-03
A10 =	-5.1634E-03	-1.4231E-02	-5.8871E-02	2.9507E-02	2.6984E-05	1.2498E-04
A12 =	1.2072E-03	6.1527E-03	3.3981E-02	-7.6788E-03	4.3984E-05	-4.6066E-06
A14 =	4.9311E-04	-7.9443E-04	-1.0582E-02	1.0697E-03	4.9484E-06	2.0601E-08
A16 =	-1.8370E-04		1.3382E-03	-6.0943E-05	-9.2829E-07	

In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 7 and Table 8 as the following values and satisfy the following conditions:

4th Embodiment			
f [mm]	4.55	(R11 - R12)/(R11 + R12)	0.19
Fno	1.90	f2/f1	-0.05
HFOV [deg.]	37.6	f/f5	-0.12
V2	55.7	f/f6	-0.29
CT6/CT4	1.19	Yc62/TL	0.30
T23/T34	0.29	SL/TL	0.97
T56/T45	0.24	TL/EPD	2.48
(R7 + R8)/(R7 - R8)	0.13	TL/lmgH	1.65

5th Embodiment

FIG. 9 is a schematic view of an image capturing device according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 5th embodiment.

In FIG. 9, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 590. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 510, an aperture stop 500, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, a sixth lens element 560, an IR-cut filter 570 and an image plane 580, wherein the image capturing lens assembly has a total of six lens elements (510-560) with refractive power.

The first lens element 510 with negative refractive power has an object-side surface 511 being convex in a paraxial

region thereof and an image-side surface 512 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 510 is made of plastic material.

The second lens element 520 with positive refractive power has an object-side surface 521 being convex in a paraxial region thereof and an image-side surface 522 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 520 is made of plastic material.

The third lens element 530 with negative refractive power has an object-side surface 531 being concave in a paraxial region thereof and an image-side surface 532 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 530 is made of plastic material.

The fourth lens element 540 with positive refractive power has an object-side surface 541 being convex in a paraxial region thereof and an image-side surface 542 being concave in a paraxial region thereof, which are both aspheric, and the fourth lens element 540 is made of plastic material.

The fifth lens element 550 with negative refractive power has an object-side surface 551 being concave in a paraxial region thereof and an image-side surface 552 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 550 is made of plastic material.

The sixth lens element 560 with positive refractive power has an object-side surface 561 being convex in a paraxial region thereof and an image-side surface 562 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 560 is made of plastic material. Specifically, both of the object-side surface 561 and the image-side surface 562 of the sixth lens element 560 have at least one inflection point.

The IR-cut filter 570 is made of glass and located between the sixth lens element 560 and the image plane 580, and will not affect the focal length of the image capturing lens assembly. The image sensor 590 is disposed on or near the image plane 580 of the image capturing lens assembly.

The detailed optical data of the 5th embodiment are shown in Table 9 and the aspheric surface data are shown in Table 10 below.

TABLE 9

5th Embodiment								
f = 3.95 mm, Fno = 2.30, HFOV = 37.5 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Lens 1	1.815	ASP	0.324	Plastic	1.544	55.9	-11.27
2		1.312	ASP	0.210				
3	Ape. Stop	Plano		-0.165				
4	Lens 2	1.279	ASP	0.705	Plastic	1.544	55.9	2.32

TABLE 9-continued

5th Embodiment f = 3.95 mm, Fno = 2.30, HFOV = 37.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
5		-83.333	ASP	0.050				
6	Lens 3	-14.034	ASP	0.265	Plastic	1.640	23.3	-5.70
7		4.961	ASP	0.243				
8	Lens 4	4.246	ASP	0.331	Plastic	1.544	55.9	9.98
9		18.925	ASP	0.463				
10	Lens 5	-2.627	ASP	0.346	Plastic	1.640	23.3	-11.86
11		-4.226	ASP	0.054				
12	Lens 6	2.019	ASP	1.057	Plastic	1.544	55.9	175.83
13		1.682	ASP	0.400				
14	IR-cut filter		Plano	0.149	Glass	1.517	64.2	—
15			Plano	0.488				
16	Image		Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 10

Aspheric Coefficients						
	Surface #					
	1	2	4	5	6	7
k =	-7.1338E-01	-3.4978E+00	-2.2081E+00	2.0000E+01	-9.0000E+01	-5.9492E+01
A4 =	-4.5615E-02	-6.0221E-02	-4.0713E-02	-1.1204E-02	-4.6098E-02	-3.2625E-03
A6 =	3.1747E-02	-2.1672E-02	3.2269E-02	7.5034E-02	6.7000E-02	9.5013E-02
A8 =	-5.8733E-02	3.5288E-02	-1.5851E-02	-4.5518E-01	-2.0294E-01	-3.3364E-01
A10 =	6.3122E-02	-1.3977E-02	-3.3801E-02	1.5169E-01	-2.3419E-01	6.5015E-01
A12 =	-3.4201E-02	-4.6957E-03	6.0074E-02	3.2030E-01	6.9624E-01	-5.8346E-01
A14 =	6.6850E-03	-3.0303E-03	-6.8581E-02	-1.9028E-01	-3.1813E-01	2.0805E-01
	Surface #					
	8	9	10	11	12	13
k =	1.4497E+01	-5.0000E+01	-3.3211E+01	-1.0000E+00	-3.5748E+01	-5.4766E+00
A4 =	-7.2181E-02	2.5804E-02	8.2418E-02	-1.3986E-01	-1.4883E-01	-6.9004E-02
A6 =	-2.0760E-01	-2.3297E-01	-3.0554E-01	4.2997E-01	4.6717E-02	2.2272E-02
A8 =	4.5730E-01	3.6894E-01	7.1652E-01	-6.8791E-01	-7.9686E-03	-5.0533E-03
A10 =	-5.5080E-01	-3.7628E-01	-1.3350E+00	5.9808E-01	4.8304E-04	5.2868E-04
A12 =	-6.6592E-02	1.6139E-01	1.4017E+00	-2.9889E-01	1.9893E-04	-1.6539E-05
A14 =	5.4050E-01	-1.4838E-02	-7.7077E-01	7.7503E-02	-1.0067E-05	-4.2936E-07
A16 =	-2.9813E-01		1.6782E-01	-7.8904E-03	-5.8574E-06	

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In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 9 and Table 10 as the following values and satisfy the following conditions:

5th Embodiment		
f [mm]	3.95 (R11 - R12)/(R11 + R12)	0.09
Fno	2.30 f2/f1	-0.21
HFOV [deg.]	37.5 f/f5	-0.33
V2	55.9 f/f6	0.02
CT6/CT4	3.19 Yc62/TL	0.30
T23/T34	0.21 SL/TL	0.89
T56/T45	0.12 TL/EPD	2.86
(R7 + R8)/(R7 - R8)	-1.58 TL/lmgH	1.59

6th Embodiment

FIG. 11 is a schematic view of an image capturing device according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 6th embodiment.

In FIG. 11, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 690. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 610, an aperture stop 600, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, a sixth lens element 660, an IR-cut filter 670 and an image plane 680, wherein the image capturing lens assembly has a total of six lens elements (610-660) with refractive power.

The first lens element 610 with negative refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 610 is made of plastic material.

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The second lens element **620** with positive refractive power has an object-side surface **621** being convex in a paraxial region thereof and an image-side surface **622** being concave in a paraxial region thereof, which are both aspheric, and the second lens element **620** is made of plastic material.

The third lens element **630** with positive refractive power has an object-side surface **631** being convex in a paraxial region thereof and an image-side surface **632** being concave in a paraxial region thereof, which are both aspheric, and the third lens element **630** is made of plastic material.

The fourth lens element **640** with positive refractive power has an object-side surface **641** being convex in a paraxial region thereof and an image-side surface **642** being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element **640** is made of plastic material.

The fifth lens element **650** with negative refractive power has an object-side surface **651** being concave in a paraxial region thereof and an image-side surface **652** being convex in

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a paraxial region thereof, which are both aspheric, and the fifth lens element **650** is made of plastic material.

The sixth lens element **660** with positive refractive power has an object-side surface **661** being convex in a paraxial region thereof and an image-side surface **662** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **660** is made of plastic material. Specifically, both of the object-side surface **661** and the image-side surface **662** of the sixth lens element **660** have at least one inflection point.

The IR-cut filter **670** is made of glass and located between the sixth lens element **660** and the image plane **680**, and will not affect the focal length of the image capturing lens assembly. The image sensor **690** is disposed on or near the image plane **680** of the image capturing lens assembly.

The detailed optical data of the 6th embodiment are shown in Table 11 and the aspheric surface data are shown in Table 12 below.

TABLE 11

6th Embodiment							
f = 3.73 mm, Fno = 2.40, HFOV = 37.2 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	3.041 ASP	0.273	Plastic	1.640	23.3	-5.96
2		1.632 ASP	0.239				
3	Ape. Stop	Plano	-0.174				
4	Lens 2	1.081 ASP	0.474	Plastic	1.535	55.7	3.04
5		2.726 ASP	0.050				
6	Lens 3	2.020 ASP	0.400	Plastic	1.632	23.4	108.90
7		1.921 ASP	0.252				
8	Lens 4	5.957 ASP	0.379	Plastic	1.544	55.9	5.60
9		-6.107 ASP	0.546				
10	Lens 5	-1.779 ASP	0.272	Plastic	1.650	21.4	-5.00
11		-4.174 ASP	0.165				
12	Lens 6	1.515 ASP	1.070	Plastic	1.535	55.7	9.72
13		1.611 ASP	0.500				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.231				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 12

Aspheric Coefficients						
	Surface #					
	1	2	4	5	6	7
k =	-8.5733E+00	-1.2959E+01	-5.1208E+00	-2.0000E+01	-3.7615E+00	-2.7572E+00
A4 =	-5.3391E-02	-7.3823E-02	1.3551E-01	5.4066E-04	-2.2298E-01	-1.2587E-01
A6 =	8.3785E-02	-2.4247E-03	-8.8183E-02	1.3646E-01	3.0813E-01	2.3947E-01
A8 =	-1.1239E-01	8.9841E-02	-5.6468E-02	-6.4369E-01	-4.5769E-01	-5.9659E-01
A10 =	1.3521E-01	2.3533E-03	2.2453E-01	-6.2114E-02	-3.2513E-01	1.0884E+00
A12 =	-7.5429E-02	-3.5548E-02	-2.6740E-01	1.0639E+00	1.2084E+00	-9.2818E-01
A14 =	1.6677E-02	1.5358E-02	1.1107E-01	-6.0303E-01	-6.2875E-01	3.0669E-01
	Surface #					
	8	9	10	11	12	13
k =	-2.0000E+01	3.8988E+00	9.7761E-01	-2.6874E-01	-1.6618E+01	-4.3167E+00
A4 =	-8.5908E-03	3.5454E-02	1.2690E-01	-3.3746E-01	-1.0988E-01	-6.7837E-02
A6 =	-1.0458E-01	-2.0406E-01	-3.4082E-01	7.8792E-01	5.3628E-02	2.5879E-02
A8 =	2.3576E-01	4.5534E-01	1.0412E+00	-1.2429E+00	-1.4011E-02	-7.0716E-03
A10 =	-3.1458E-01	-6.5948E-01	-2.5039E+00	1.2128E+00	1.2121E-03	1.1966E-03
A12 =	2.2372E-02	4.6423E-01	3.3216E+00	-7.2187E-01	3.0964E-04	-1.1998E-04
A14 =	3.7973E-01	-1.0314E-01	-2.3089E+00	2.3533E-01	-8.0118E-05	5.2951E-06
A16 =	-2.4653E-01		6.4598E-01	-3.1457E-02	5.1703E-06	

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In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 11 and Table 12 as the following values and satisfy the following conditions:

6th Embodiment			
f [mm]	3.73	$(R11 - R12)/(R11 + R12)$	-0.03
Fno	2.40	f2/f1	-0.51
HFOV [deg.]	37.2	f/f5	-0.75
V2	55.7	f/f6	0.38
CT6/CT4	2.82	Yc62/TL	0.36
T23/T34	0.20	SL/TL	0.90
T56/T45	0.30	TL/EPD	3.13
$(R7 + R8)/(R7 - R8)$	-0.01	TL/lmgH	1.71

7th Embodiment

FIG. 13 is a schematic view of an image capturing device according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 7th embodiment.

In FIG. 13, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 790. The image capturing lens assembly includes, in order from an object side to an image side, a first lens element 710, a second lens element 720, an aperture stop 700, a third lens element 730, a fourth lens element 740, a fifth lens element 750, a sixth lens element 760, an IR-cut filter 770 and an image plane 780, wherein the image capturing lens assembly has a total of six lens elements (710-760) with refractive power.

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The first lens element 710 with negative refractive power has an object-side surface 711 being convex in a paraxial region thereof and an image-side surface 712 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 710 is made of plastic material.

The second lens element 720 with positive refractive power has an object-side surface 721 being convex in a paraxial region thereof and an image-side surface 722 being convex in a paraxial region thereof, which are both aspheric, and the second lens element 720 is made of plastic material.

The third lens element 730 with negative refractive power has an object-side surface 731 being convex in a paraxial region thereof and an image-side surface 732 being concave in a paraxial region thereof, which are both aspheric, and the third lens element 730 is made of plastic material.

The fourth lens element 740 with negative refractive power has an object-side surface 741 being concave in a paraxial region thereof and an image-side surface 742 being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element 740 is made of plastic material.

The fifth lens element 750 with positive refractive power has an object-side surface 751 being concave in a paraxial region thereof and an image-side surface 752 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 750 is made of plastic material.

The sixth lens element 760 with negative refractive power has an object-side surface 761 being convex in a paraxial region thereof and an image-side surface 762 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 760 is made of plastic material. Moreover, both of the object-side surface 761 and the image-side surface 762 of the sixth lens element 760 have at least one inflection point.

The IR-cut filter 770 is made of glass and located between the sixth lens element 760 and the image plane 780, and will not affect the focal length of the image capturing lens assembly. The image sensor 790 is disposed on or near the image plane 780 of the image capturing lens assembly.

The detailed optical data of the 7th embodiment are shown in Table 13 and the aspheric surface data are shown in Table 14 below.

TABLE 13

7th Embodiment							
f = 4.70 mm, Fno = 2.70, HFOV = 34.5 deg.							
Surface #		Curvature	Radius	Thickness	Material	Index	Focal Length
0	Object	Plano		Infinity			
1	Lens 1	2.084	ASP	0.411	Plastic	1.535	-25.84
2		1.686	ASP	0.067			
3	Lens 2	1.592	ASP	0.452	Plastic	1.544	2.69
4		-16.021	ASP	-0.010			
5	Ape. Stop	Plano		0.060			
6	Lens 3	7.759	ASP	0.377	Plastic	1.639	-6.97
7		2.776	ASP	0.420			
8	Lens 4	-31.936	ASP	0.506	Plastic	1.544	-81.66
9		-114.169	ASP	0.384			
10	Lens 5	-6.178	ASP	0.374	Plastic	1.639	108.79
11		-5.808	ASP	0.135			
12	Lens 6	2.093	ASP	0.790	Plastic	1.535	-24.55
13		1.568	ASP	0.500			
14	IR-cut filter	Plano		0.165	Glass	1.517	—
15		Plano		0.820			
16	Image	Plano		—			

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients						
Surface #						
	1	2	3	4	6	7
k =	-1.1145E+00	-3.5628E+00	-2.0635E+00	1.0175E+01	1.7133E+01	-1.3180E+01
A4 =	-3.7167E-02	-8.5356E-02	-4.2914E-02	-3.3575E-02	-7.4024E-02	3.7735E-02
A6 =	8.3642E-03	-4.2018E-02	-2.2409E-03	8.6693E-02	9.3178E-02	1.9491E-02
A8 =	-3.3719E-02	6.3907E-02	3.0791E-02	-1.7313E-01	-1.1151E-01	-1.2417E-01
A10 =	2.8105E-02	1.7479E-02	2.3053E-02	1.1305E-01	-8.8224E-02	2.5485E-01
A12 =	-8.0420E-03	-1.3598E-02	1.0001E-02	7.3364E-02	3.4129E-01	-2.7488E-01
A14 =	-6.1212E-04	-1.2695E-02	-8.1027E-03	-5.5484E-02	-2.1230E-01	1.2231E-01
Surface #						
	8	9	10	11	12	13
k =	-5.3312E+01	-1.0765E+01	-1.9273E+00	-1.0000E+00	-2.6866E+01	-7.5011E+00
A4 =	-4.4613E-02	-2.8018E-02	1.4090E-01	-8.2960E-02	-1.4181E-01	-6.8494E-02
A6 =	-3.3446E-02	-7.1253E-02	-3.0997E-01	2.2145E-01	4.1901E-02	1.7058E-02
A8 =	1.4163E-01	1.4059E-01	4.3693E-01	-3.2253E-01	-4.0994E-03	-2.7540E-03
A10 =	-1.7925E-01	-1.3084E-01	-5.5508E-01	2.3357E-01	-6.5174E-05	1.6847E-04
A12 =	2.0693E-02	6.6507E-02	4.4014E-01	-9.3945E-02	3.3178E-05	3.0197E-06
A14 =	1.1497E-01	-1.3704E-02	-1.8921E-01	1.9522E-02	-1.2589E-06	-4.4099E-07
A16 =	-7.2897E-02		3.2986E-02	-1.5741E-03	-2.2145E-09	

In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 13 and Table 14 as the following values and satisfy the following conditions:

7th Embodiment			
f [mm]	4.70	(R11 - R12)/(R11 + R12)	0.14
Fno	2.70	f2/f1	-0.10
HFOV [deg.]	34.5	f/f5	0.04
V2	55.9	f/f6	-0.19
CT6/CT4	1.56	Yc62/TL	0.24
T23/T34	0.12	SL/TL	0.83
T56/T45	0.35	TL/EPD	3.13
(R7 + R8)/(R7 - R8)	-1.78	TL/lmgH	1.65

8th Embodiment

FIG. 15 is a schematic view of an image capturing device according to the 8th embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing device according to the 8th embodiment.

In FIG. 15, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor **890**. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop **800**, a first lens element **810**, a second lens element **820**, a third lens element **830**, a fourth lens element **840**, a fifth lens element **850**, a sixth lens element **860**, an IR-cut filter **870** and an image plane **880**, wherein the image capturing lens assembly has a total of six lens elements (**810-860**) with refractive power.

The first lens element **810** with negative refractive power has an object-side surface **811** being convex in a paraxial

region thereof and an image-side surface **812** being concave in a paraxial region thereof, which are both aspheric, and the first lens element **810** is made of plastic material.

The second lens element **820** with positive refractive power has an object-side surface **821** being convex in a paraxial region thereof and an image-side surface **822** being concave in a paraxial region thereof, which are both aspheric, and the second lens element **820** is made of plastic material.

The third lens element **830** with positive refractive power has an object-side surface **831** being convex in a paraxial region thereof and an image-side surface **832** being convex in a paraxial region thereof, which are both aspheric, and the third lens element **830** is made of plastic material.

The fourth lens element **840** with negative refractive power has an object-side surface **841** being concave in a paraxial region thereof and an image-side surface **842** being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element **840** is made of plastic material.

The fifth lens element **850** with negative refractive power has an object-side surface **851** being concave in a paraxial region thereof and an image-side surface **852** being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element **850** is made of plastic material.

The sixth lens element **860** with positive refractive power has an object-side surface **861** being convex in a paraxial region thereof and an image-side surface **862** being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element **860** is made of plastic material. Moreover, both of the object-side surface **861** and the image-side surface **862** of the sixth lens element **860** have at least one inflection point.

The IR-cut filter **870** is made of glass and located between the sixth lens element **860** and the image plane **880**, and will not affect the focal length of the image capturing lens assembly. The image sensor **890** is disposed on or near the image plane **880** of the image capturing lens assembly.

The detailed optical data of the 8th embodiment are shown in Table 15 and the aspheric surface data are shown in Table 16 below.

TABLE 15

8th Embodiment f = 3.44 mm, Fno = 2.25, HFOV = 38.9 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.087				
2	Lens 1	2.197 ASP	0.240	Plastic	1.535	56.3	-171.55
3		2.064 ASP	0.101				
4	Lens 2	1.340 ASP	0.338	Plastic	1.535	56.3	8.76
5		1.713 ASP	0.156				
6	Lens 3	4.521 ASP	0.623	Plastic	1.544	55.9	2.58
7		-1.942 ASP	0.264				
8	Lens 4	-0.996 ASP	0.579	Plastic	1.650	21.4	-3.91
9		-2.015 ASP	0.176				
10	Lens 5	-15.574 ASP	0.240	Plastic	1.544	55.9	-95.83
11		-22.326 ASP	0.050				
12	Lens 6	1.292 ASP	0.778	Plastic	1.535	56.3	12.20
13		1.274 ASP	0.600				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.554				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients						
	Surface #					
	2	3	4	5	6	7
k =	-1.8575E+01	-5.1303E+00	-4.5364E-01	5.6417E-01	3.0000E+00	-1.0473E+00
A4 =	9.9371E-02	-4.4448E-01	-5.3097E-01	-2.8595E-01	-1.6789E-01	-1.5652E-01
A6 =	1.5215E-02	1.0979E+00	8.4647E-01	3.7353E-02	-8.8489E-02	-5.9621E-02
A8 =	-4.8966E-01	-2.3240E+00	-1.6581E+00	1.8581E-01	9.9012E-02	1.4702E-01
A10 =	7.9182E-01	2.3304E+00	1.6951E+00	-7.5398E-01	-6.0571E-02	-1.2411E-01
A12 =	-6.1282E-01	-1.1007E+00	-1.2147E+00	6.6732E-01	-2.1203E-01	-1.1167E-02
A14 =			3.4122E-01	-2.2440E-01	1.7337E-01	4.0901E-02
	Surface #					
	8	9	10	11	12	13
k =	-1.4128E+00	-2.2081E+00	3.0002E+00	-5.2607E-01	-3.6812E+00	-1.4198E+00
A4 =	-9.1222E-02	-1.6495E-01	2.0776E-01	2.7687E-01	-1.1561E-01	-1.9955E-01
A6 =	6.9053E-02	3.0748E-01	-2.4506E-01	-3.6249E-01	2.7445E-02	1.0624E-01
A8 =	4.0401E-01	-2.4096E-01	1.2850E-01	3.0165E-01	1.5008E-02	-4.3883E-02
A10 =	-8.2890E-01	1.0791E-01	-3.8174E-02	-1.7269E-01	-1.0621E-02	1.2870E-02
A12 =	6.1161E-01	-3.5191E-02	-5.1522E-03	6.0617E-02	2.3500E-03	-2.4187E-03
A14 =	-1.5475E-01	8.2714E-03	3.9680E-03	-1.1466E-02	-1.7812E-04	2.5402E-04
A16 =				8.9190E-04	-1.2558E-06	-1.1268E-05

In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 15 and Table 16 as the following values and satisfy the following conditions:

8th Embodiment			
f [mm]	3.44	(R11 - R12)/(R11 + R12)	0.01
Fno	2.25	f2/f1	-0.05

-continued

8th Embodiment			
55	HFOV [deg.]	38.9	f/f5
	V2	56.3	f/f6
	CT6/CT4	1.34	Yc62/TL
	T23/T34	0.59	SL/TL
	T56/T45	0.28	TL/EPD
60	(R7 + R8)/(R7 - R8)	-2.96	TL/lmgH
			1.72

9th Embodiment

65 FIG. 17 is a schematic view of an image capturing device according to the 9th embodiment of the present disclosure. FIG. 18 shows, in order from left to right, spherical aberration

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curves, astigmatic field curves and a distortion curve of the image capturing device according to the 9th embodiment.

In FIG. 17, the image capturing device includes the image capturing lens assembly (not otherwise herein labeled) of the present disclosure and an image sensor 990. The image capturing lens assembly includes, in order from an object side to an image side, an aperture stop 900, a first lens element 910, a second lens element 920, a third lens element 930, a fourth lens element 940, a fifth lens element 950, a sixth lens element 960, an IR-cut filter 970 and an image plane 980, wherein the image capturing lens assembly has a total of six lens elements (910-960) with refractive power.

The first lens element 910 with negative refractive power has an object-side 911 surface being convex in a paraxial region thereof and an image-side surface 912 being concave in a paraxial region thereof, which are both aspheric, and the first lens element 910 is made of plastic material.

The second lens element 920 with positive refractive power has an object-side surface 921 being convex in a paraxial region thereof and an image-side surface 922 being concave in a paraxial region thereof, which are both aspheric, and the second lens element 920 is made of plastic material.

The third lens element 930 with positive refractive power has an object-side surface 931 being convex in a paraxial region thereof and an image-side surface 932 being convex in a paraxial region thereof, which are both aspheric, and the third lens element 930 is made of plastic material.

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The fourth lens element 940 with negative refractive power has an object-side surface 941 being concave in a paraxial region thereof and an image-side surface 942 being convex in a paraxial region thereof, which are both aspheric, and the fourth lens element 940 is made of plastic material.

The fifth lens element 950 with positive refractive power has an object-side surface 951 being concave in a paraxial region thereof and an image-side surface 952 being convex in a paraxial region thereof, which are both aspheric, and the fifth lens element 950 is made of plastic material.

The sixth lens element 960 with positive refractive power has an object-side surface 961 being convex in a paraxial region thereof and an image-side surface 962 being concave in a paraxial region thereof, which are both aspheric, and the sixth lens element 960 is made of plastic material. Moreover, both of the object-side surface 961 and the image-side surface 962 of the sixth lens element 960 have at least one inflection point.

The IR-cut filter 970 is made of glass and located between the sixth lens element 960 and the image plane 980, and will not affect the focal length of the image capturing lens assembly. The image sensor 990 is disposed on the image plane 980 of the image capturing lens assembly.

The detailed optical data of the 9th embodiment are shown in Table 17 and the aspheric surface data are shown in Table 18 below.

TABLE 17

9th Embodiment							
f = 3.06 mm, Fno = 2.30, HFOV = 42.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.040				
2	Lens 1	3.139 ASP	0.240	Plastic	1.650	21.4	-19.34
3		2.436 ASP	0.114				
4	Lens 2	1.271 ASP	0.384	Plastic	1.530	55.8	8.77
5		1.567 ASP	0.123				
6	Lens 3	2.308 ASP	0.667	Plastic	1.544	55.9	2.53
7		-3.069 ASP	0.202				
8	Lens 4	-1.266 ASP	0.400	Plastic	1.650	21.4	-3.93
9		-2.824 ASP	0.183				
10	Lens 5	-11.571 ASP	0.336	Plastic	1.544	55.9	16.83
11		-5.165 ASP	0.046				
12	Lens 6	1.353 ASP	0.893	Plastic	1.535	56.3	19.32
13		1.198 ASP	0.600				
14	IR-cut filter	Plano	0.200	Glass	1.517	64.2	—
15		Plano	0.260				
16	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 18

Aspheric Coefficients						
Surface #						
	2	3	4	5	6	7
k =	-4.8065E+01	9.0894E-01	3.5877E-02	9.0263E-01	2.3933E+00	-9.9998E+00
A4 =	7.0153E-02	-3.7741E-01	-4.1871E-01	-2.6107E-01	-1.4488E-01	-7.1397E-02
A6 =	-8.2562E-03	9.6928E-01	8.4618E-01	1.1096E-01	-4.6178E-02	-7.7573E-02
A8 =	-4.3612E-01	-2.2098E+00	-1.6252E+00	2.1091E-01	1.3007E-01	1.2838E-01
A10 =	8.5472E-01	2.6147E+00	1.7756E+00	-7.6661E-01	-3.5857E-02	-1.3110E-01
A12 =	-5.8773E-01	-1.2616E+00	-1.1050E+00	6.4258E-01	-2.1688E-01	-9.5546E-03
A14 =			2.7489E-01	-2.1860E-01	1.1916E-01	3.8788E-02

TABLE 18-continued

	Aspheric Coefficients					
	Surface #					
	8	9	10	11	12	13
k =	-2.1754E+00	-6.9654E+00	2.9283E+00	-6.5975E+00	-2.1816E+00	-1.4122E+00
A4 =	-7.4036E-02	-1.5896E-01	3.1280E-01	3.7231E-01	-1.5105E-01	-1.9639E-01
A6 =	5.1894E-02	2.9655E-01	-2.9545E-01	-3.8980E-01	2.0192E-02	1.0473E-01
A8 =	3.8089E-01	-2.4334E-01	1.4464E-01	2.9985E-01	1.4701E-02	-4.4293E-02
A10 =	-8.4732E-01	1.1172E-01	-3.5169E-02	-1.7164E-01	-1.0079E-02	1.2888E-02
A12 =	6.1184E-01	-3.2955E-02	-6.3347E-03	6.0813E-02	2.6680E-03	-2.3993E-03
A14 =	-1.4223E-01	7.6813E-03	3.3718E-03	-1.1469E-02	-1.2767E-04	2.5391E-04
A16 =				8.7537E-04	-3.8629E-05	-1.1530E-05

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In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

Moreover, these parameters can be calculated from Table 17 and Table 18 as the following values and satisfy the following conditions:

9th Embodiment			
f [mm]	3.06	(R11 - R12)/(R11 + R12)	0.06
Fno	2.30	f2/f1	-0.45
HFOV [deg.]	42.0	f/f5	0.18
V2	55.8	f/f6	0.16
CT6/CT4	2.23	Yc62/TL	0.36
T23/T34	0.61	SL/TL	0.99
T56/T45	0.25	TL/EPD	3.49
(R7 + R8)/(R7 - R8)	-2.62	TL/ImgH	1.63

The foregoing image capturing device **10** may be installed in but not limited to a mobile terminal, including a smart phone (FIG. **20**), a tablet personal computer (FIG. **21**) or a wearable device (FIG. **22**). Since the present disclosure of the first lens element has negative refractive power, it is favorable for enlarging the field of view so as to capture more of the image scene for the mobile terminal. Moreover, the second lens element has the major light convergence ability; it is favorable for effectively reducing the back focal length of the image capturing lens assembly so as to reduce the total track length thereof. In addition, it is also favorable for avoiding the refractive powers in the rear of image capturing lens assembly from excessively strong resulting in excessive aberration and worse image quality.

The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-18 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical applications, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An image capturing lens assembly comprising, in order from an object side to an image side:

a first lens element with negative refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof;

a second lens element having positive refractive power;

a third lens element having refractive power;

a fourth lens element having refractive power;

a fifth lens element having negative refractive power, wherein an object-side surface and an image-side surface of the fifth lens element are aspheric; and

a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein both of an object-side surface and the image-side surface of the sixth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the sixth lens element has at least one inflection point;

wherein the image capturing lens assembly has a total of six lens elements with refractive power, a focal length of the image capturing lens assembly is f, a focal length of the first lens element is f1, a focal length of the second lens element is f2, a focal length of the fifth lens element is f5, and the following conditions are satisfied:

$$-1.0 < f2/f1 < 0; \text{ and}$$

$$-1.5 < f/f5 < 0.$$

2. The image capturing lens assembly of claim **1**, further comprising:

a stop, wherein an axial distance between the stop and an image plane is SL, and an axial distance between the object-side surface of the first lens element and the image plane is TL, the following condition is satisfied:

$$0.80 < SL/TL < 1.10.$$

3. The image capturing lens assembly of claim **1**, wherein an axial distance between the object-side surface of the first lens element and an image plane is TL, a maximum image height of the image capturing lens assembly is ImgH, and the following condition is satisfied:

$$TL/ImgH < 1.90.$$

4. The image capturing lens assembly of claim **1**, wherein of the first lens element, the second lens element, the third lens element, the fourth lens element, the fifth lens element and the sixth lens element of the image capturing lens assembly, there is a space in a paraxial region between every pair of lens elements that are adjacent to each other.

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5. The image capturing lens assembly of claim 1, wherein the second lens element has an object-side surface being convex in a paraxial region thereof.

6. The image capturing lens assembly of claim 5, wherein the focal length of the first lens element is f_1 , the focal length of the second lens element is f_2 , and the following condition is satisfied:

$$-0.6 < f_2/f_1 < 0.$$

7. The image capturing lens assembly of claim 5, wherein the object-side surface of the fifth lens element is concave in a paraxial region thereof.

8. The image capturing lens assembly of claim 5, wherein the focal length of the image capturing lens assembly is f , the focal length of the fifth lens element is f_5 , and the following condition is satisfied:

$$-0.85 < f/f_5 < 0.$$

9. The image capturing lens assembly of claim 2, wherein the third lens element has negative refractive power.

10. The image capturing lens assembly of claim 1, wherein the object-side surface of the sixth lens element is convex in a paraxial region thereof.

11. The image capturing lens assembly of claim 10, wherein the focal length of the image capturing lens assembly is f , a focal length of the sixth lens element is f_6 , and the following condition is satisfied:

$$-0.5 < f/f_6 < 0.5.$$

12. The image capturing lens assembly of claim 10 wherein a central thickness of the fourth lens element is CT_4 , a central thickness of the sixth lens element is CT_6 , and the following condition is satisfied:

$$1.0 < CT_6/CT_4 < 4.0.$$

13. The image capturing lens assembly of claim 8, wherein the fourth lens element has positive refractive power.

14. The image capturing lens assembly of claim 10, wherein the third lens element has an image-side surface being concave in a paraxial region thereof.

15. An image capturing lens assembly comprising, in order from an object side to an image side:

- a first lens element having negative refractive power;
- a second lens element having positive refractive power;
- a third lens element having refractive power;
- a fourth lens element having refractive power, wherein an object-side surface and an image-side surface of the fourth lens element are aspheric;
- a fifth lens element having refractive power, wherein an object-side surface and an image-side surface of the fifth lens element are aspheric; and
- a sixth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein both of an object-side surface and the image-side surface of the sixth lens element are aspheric, and at least one of the object-side surface and the image-side surface of the sixth lens element has at least one inflection point;

wherein the image capturing lens assembly has a total of six lens elements with refractive power, a focal length of the image capturing lens assembly is f , a focal length of the first lens element is f_1 , a focal length of the second lens element is f_2 , a focal length of the fifth lens element is f_5 , and an axial distance between an object-side surface of the first lens element and an image plane is TL , an

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entrance pupil diameter of the image capturing lens assembly is EPD , and the following conditions are satisfied:

$$-3.0 < f_2/f_1 < 0;$$

$$-1.5 < f/f_5 < 0.4; \text{ and}$$

$$TL/EPD < 4.0.$$

16. The image capturing lens assembly of claim 15, wherein a curvature radius of the object-side surface of the sixth lens element is R_{11} , a curvature radius of the image-side surface of the sixth lens element is R_{12} , and the following condition is satisfied:

$$-0.2 < (R_{11} - R_{12}) / (R_{11} + R_{12}) < 0.4.$$

17. The image capturing lens assembly of claim 15, wherein a vertical distance between a non-axial critical point on the image-side surface of the sixth lens element and an optical axis is Y_{c62} , the axial distance between the object-side surface of the first lens element and the image plane is TL , and the following condition is satisfied:

$$0.1 < Y_{c62}/TL < 0.7.$$

18. The image capturing lens assembly of claim 15, wherein the focal length of the first lens element is f_1 , the focal length of the second lens element is f_2 , and the following condition is satisfied:

$$-1.0 < f_2/f_1 < 0.$$

19. The image capturing lens assembly of claim 18, wherein an axial distance between the fourth lens element and the fifth lens element is T_{45} , an axial distance between the fifth lens element and the sixth lens element is T_{56} , and the following condition is satisfied:

$$0 < T_{56}/T_{45} < 0.90.$$

20. An image capturing device, comprising:
the image capturing lens assembly of claim 15; and
an image sensor, wherein the image sensor is located on an image side of the image capturing lens assembly.

21. A mobile terminal, comprising:
the image capturing device of claim 20.

22. The image capturing lens assembly of claim 15, wherein the first lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof.

23. The image capturing lens assembly of claim 22, wherein an Abbe number of the second lens element is V_2 , and the following condition is satisfied:

$$40 < V_2 < 70.$$

24. The image capturing lens assembly of claim 22, wherein an axial distance between the second lens element and the third lens element is T_{23} , an axial distance between the third lens element and the fourth lens element is T_{34} , and the following condition is satisfied:

$$0 < T_{23}/T_{34} < 0.50.$$

25. The image capturing lens assembly of claim 22, wherein a curvature radius of an object-side surface of the fourth lens element is R_7 , a curvature radius of an image-side surface of the fourth lens element is R_8 , and the following condition is satisfied:

$$-0.5 < (R_7 + R_8) / (R_7 - R_8) < 0.5.$$

* * * * *

UNITED STATES PATENT AND TRADEMARK OFFICE
CERTIFICATE OF CORRECTION

PATENT NO. : 9,201,216 B2
APPLICATION NO. : 14/222267
DATED : December 1, 2015
INVENTOR(S) : Hsin-Hsuan Huang et al.

Page 1 of 1

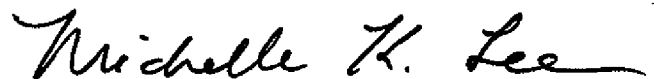
It is certified that error appears in the above-identified patent and that said Letters Patent is hereby corrected as shown below:

Claims

On Column 35, claim 9, line 1: replace "claim 2" with --claim 5--

On Column 35, claim 13, line 1: replace "claim 8" with --claim 10--

Signed and Sealed this
Fifteenth Day of March, 2016

A handwritten signature in black ink, reading "Michelle K. Lee". The signature is fluid and cursive, with the first letters of each word being capitalized and prominent.

Michelle K. Lee
Director of the United States Patent and Trademark Office